Nuclear energy

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Outline

- Short background/status
- The physics
- Nuclear reactions
- Utilization of nuclear energy/reactor types
- Waste handling
- Future directions gen. IV reactors, Fusion

Nuclear energy

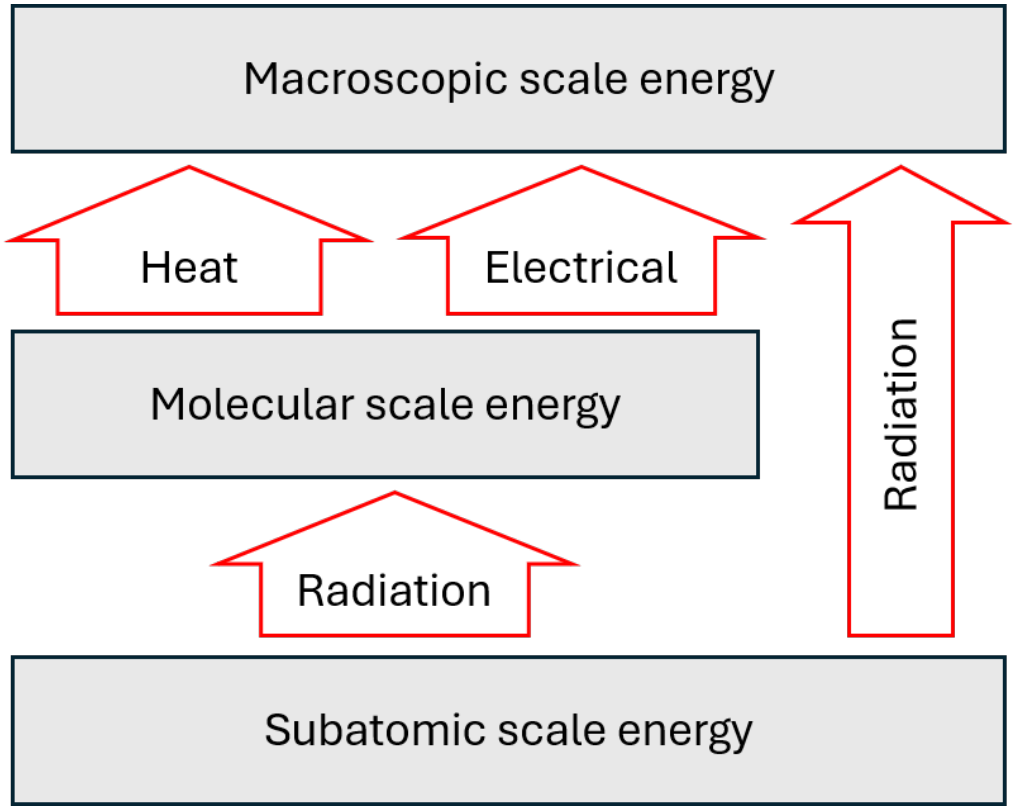
A source of energy based on the conversion of matter into energy

Based on Einsteins equation for the equivalence between energy and matter

$$E = mc^2$$

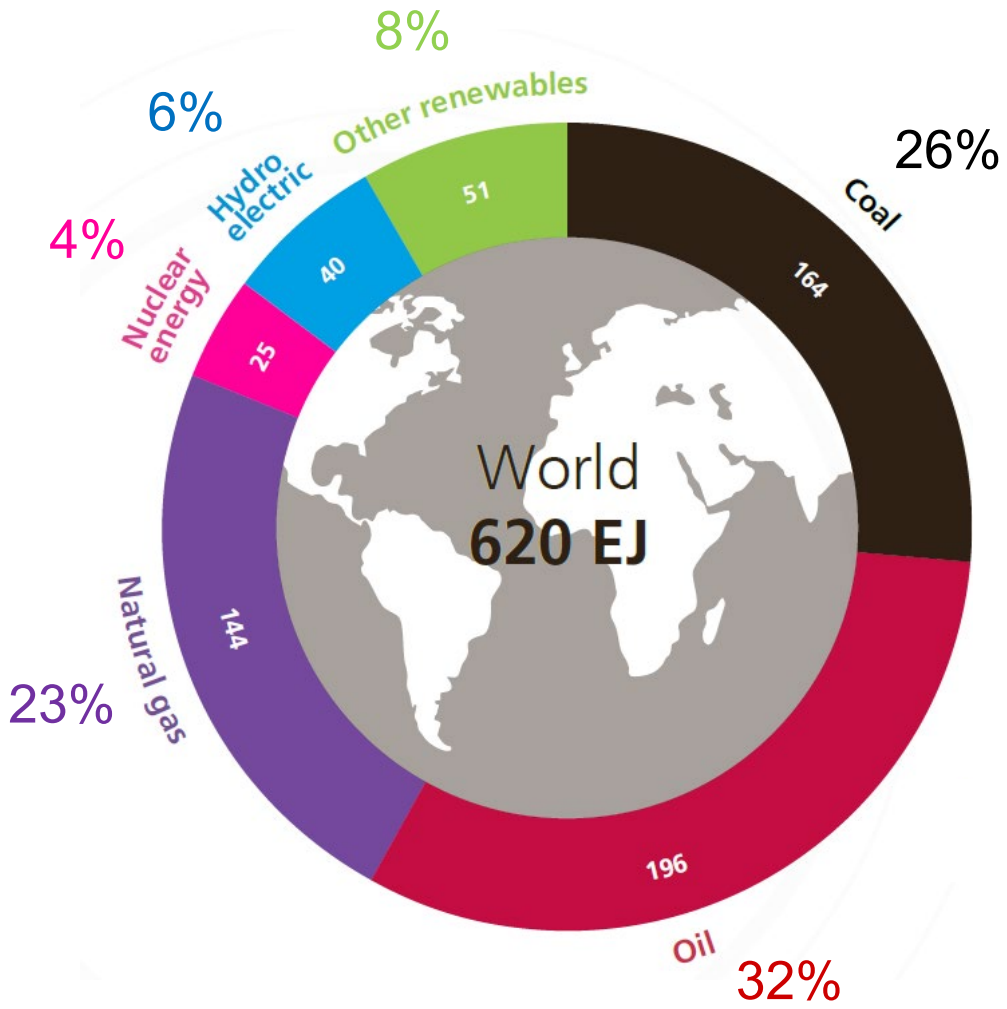
It is in the sub-atomic energy regime

Energy is harvested in the form of heat



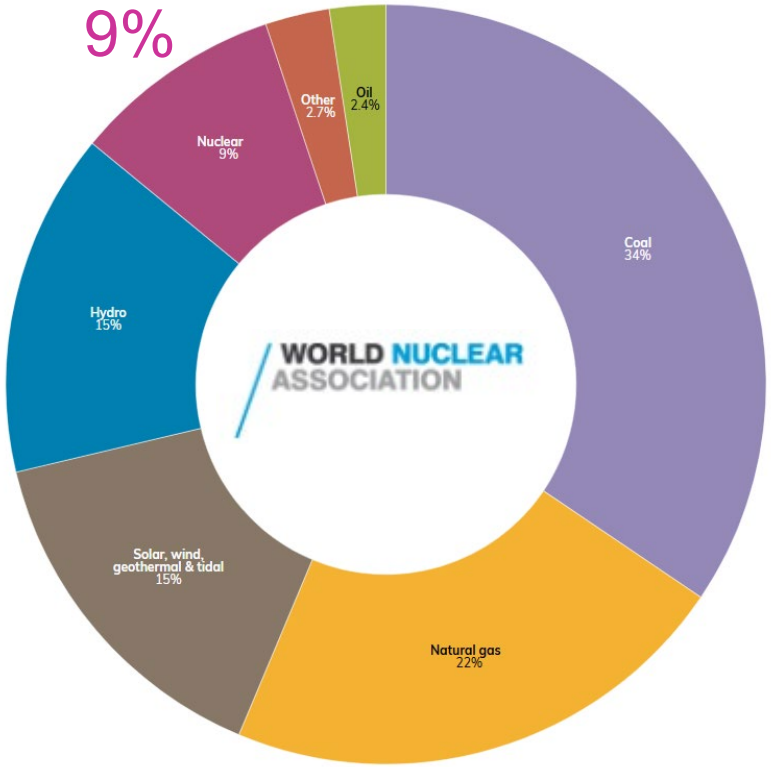
Global energy mix

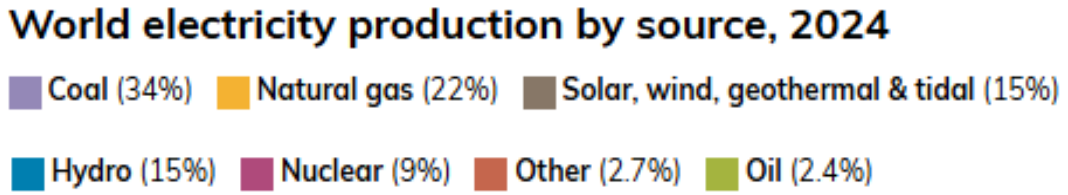
World energy production by source 2023



https://climatediscussionnexus.com/

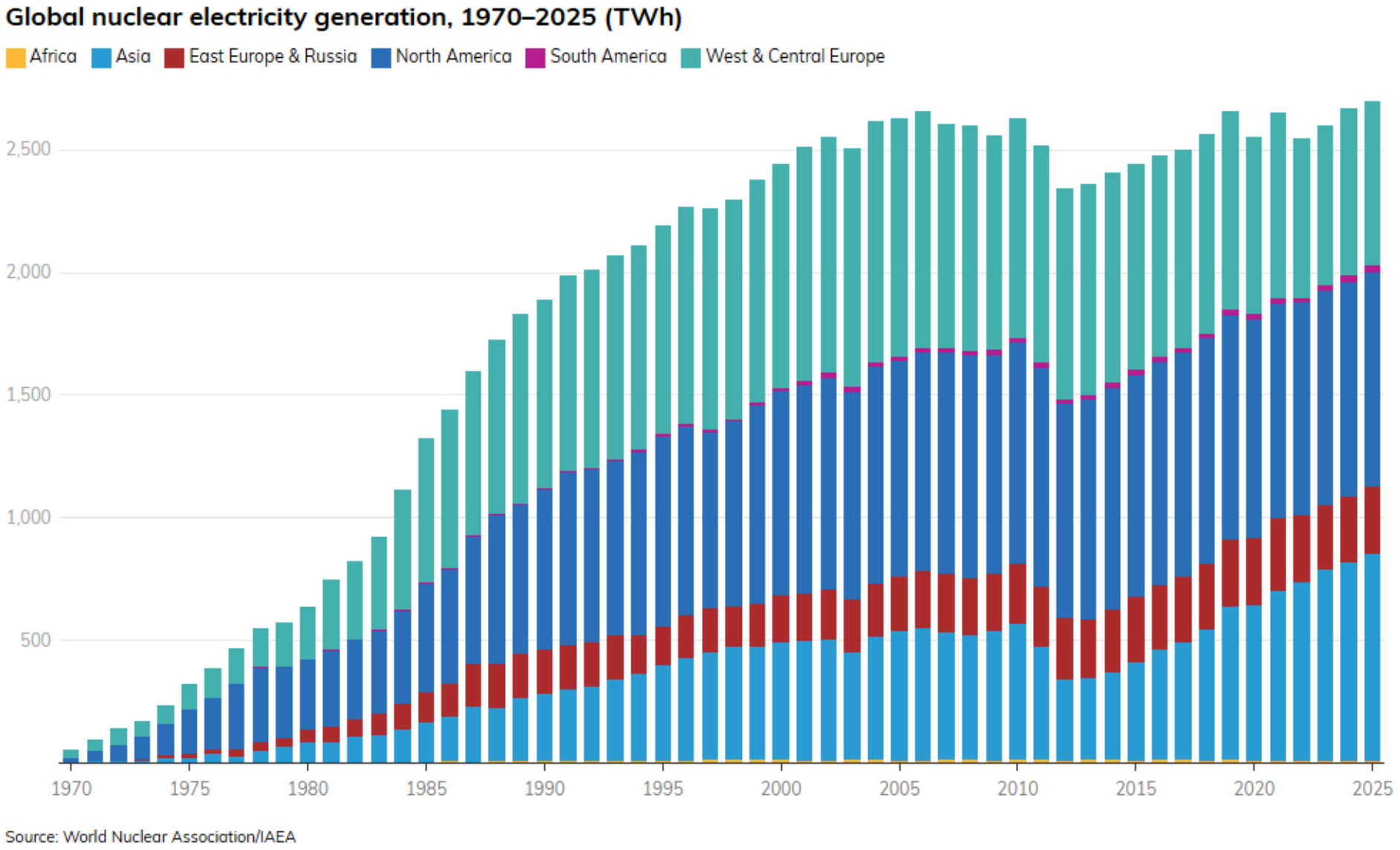
World electricity production by source 2024





https://world-nuclear.org/information-library/current-and-future-generation/nuclear-power-in-the-world-today

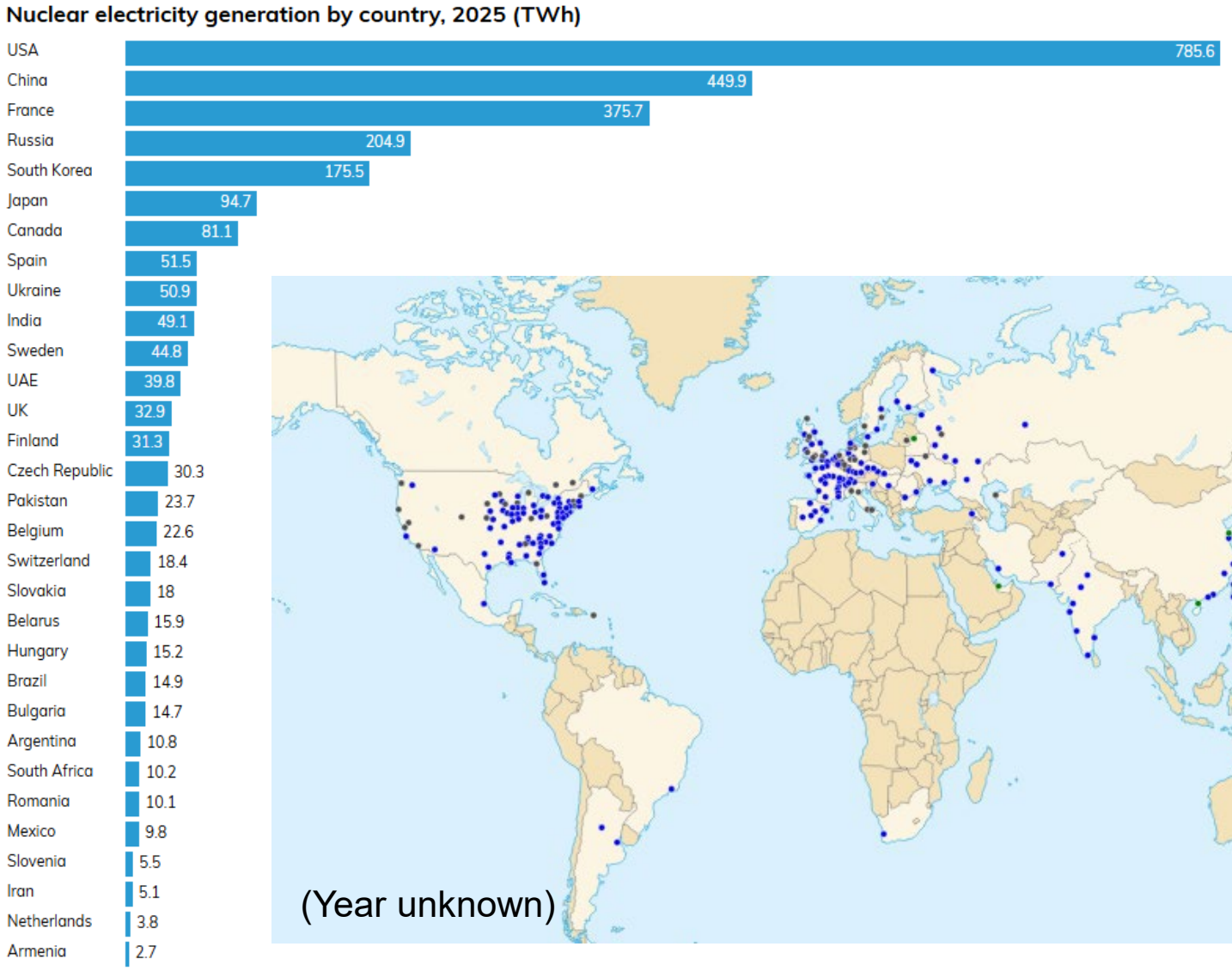
Development



https://world-nuclear.org/information-library/current-and-future-generation/nuclear-power-in-the-world-today

Which countries

https://world-nuclear.org/information-library/current-and-future-generation/nuclear-power-in-the-world-today





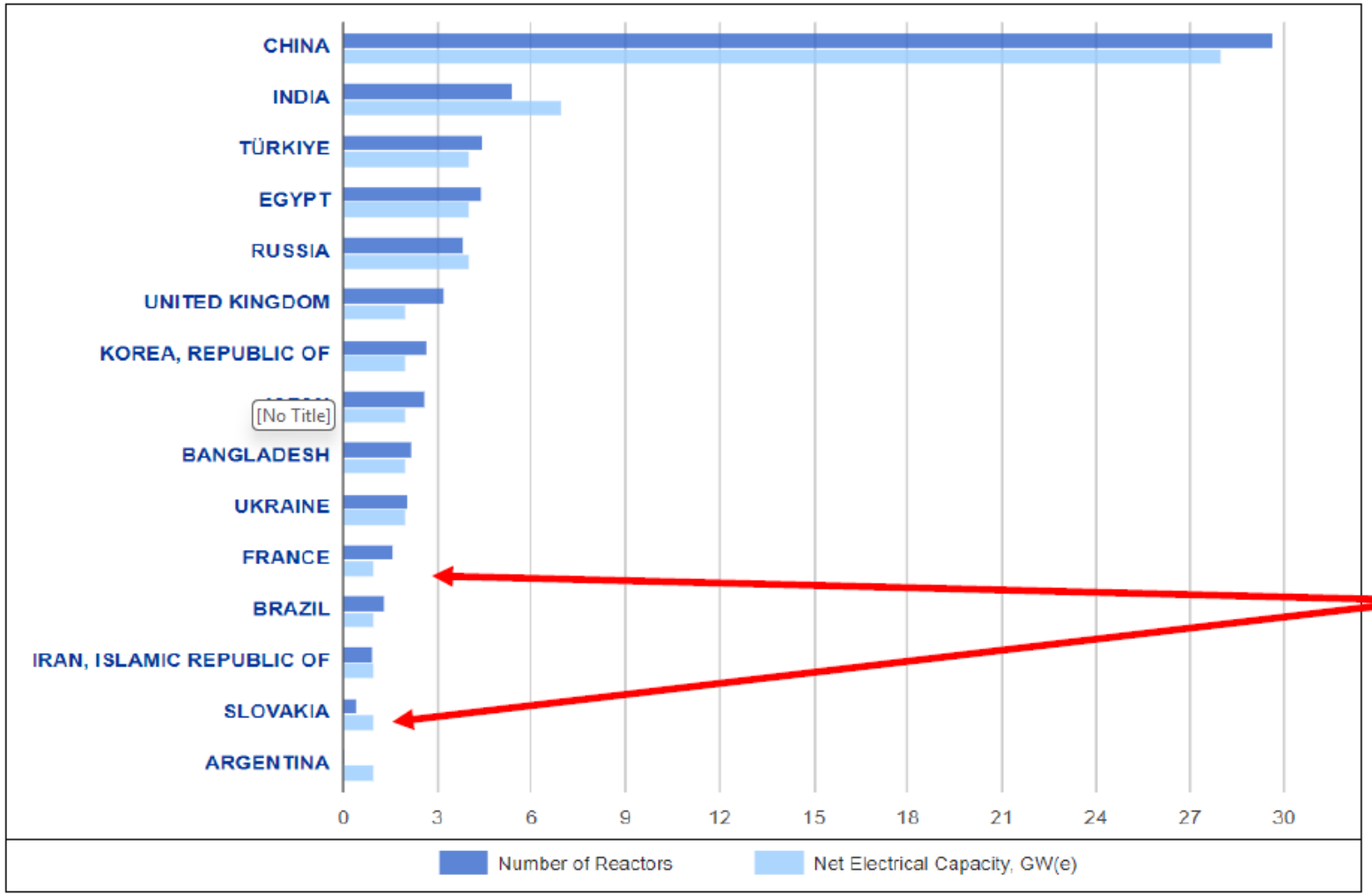
June 2926, Nuklear Newswire.
Source: EIA, with additional data from World Bank, Global Energy Monitor, Global Nuclear Power Tracker, and the IAEA. Image: EIA)

China: 60 operational reactors, and 36 under construction.

Bent Lauritzen

NPPs under construction (2024)





62 reactors, 64.9 GW_e under construction

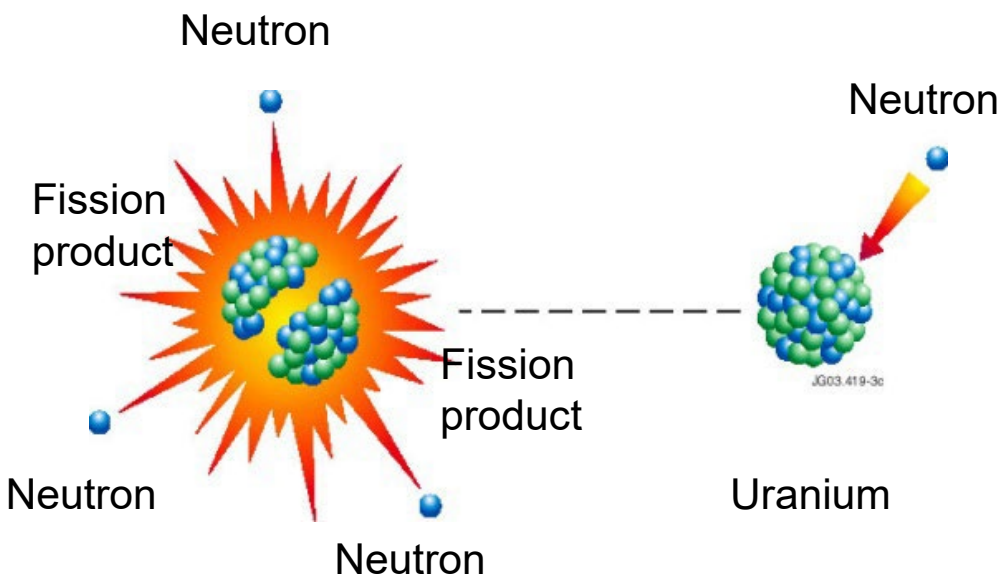
Only 2 in the EU

Only 1 European manufacturer

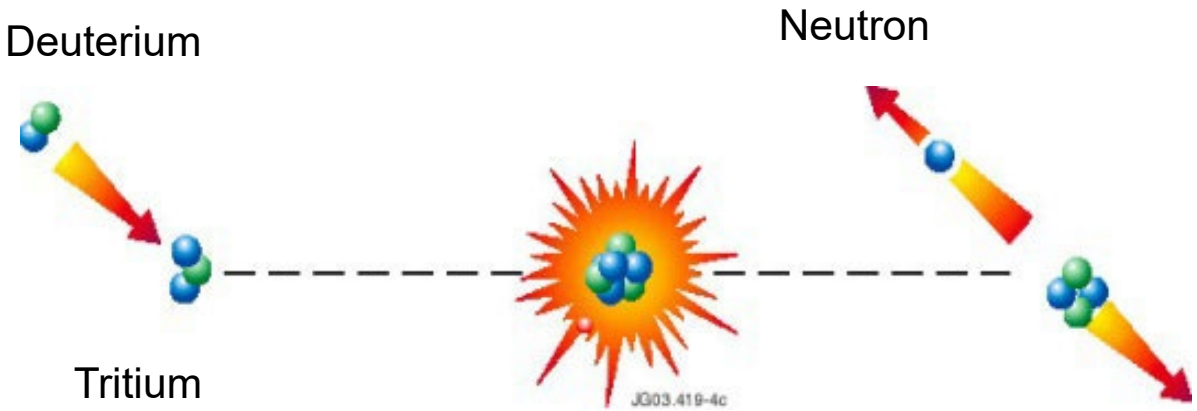
Source: 2022 IAEA

Physics

Fission and fusion

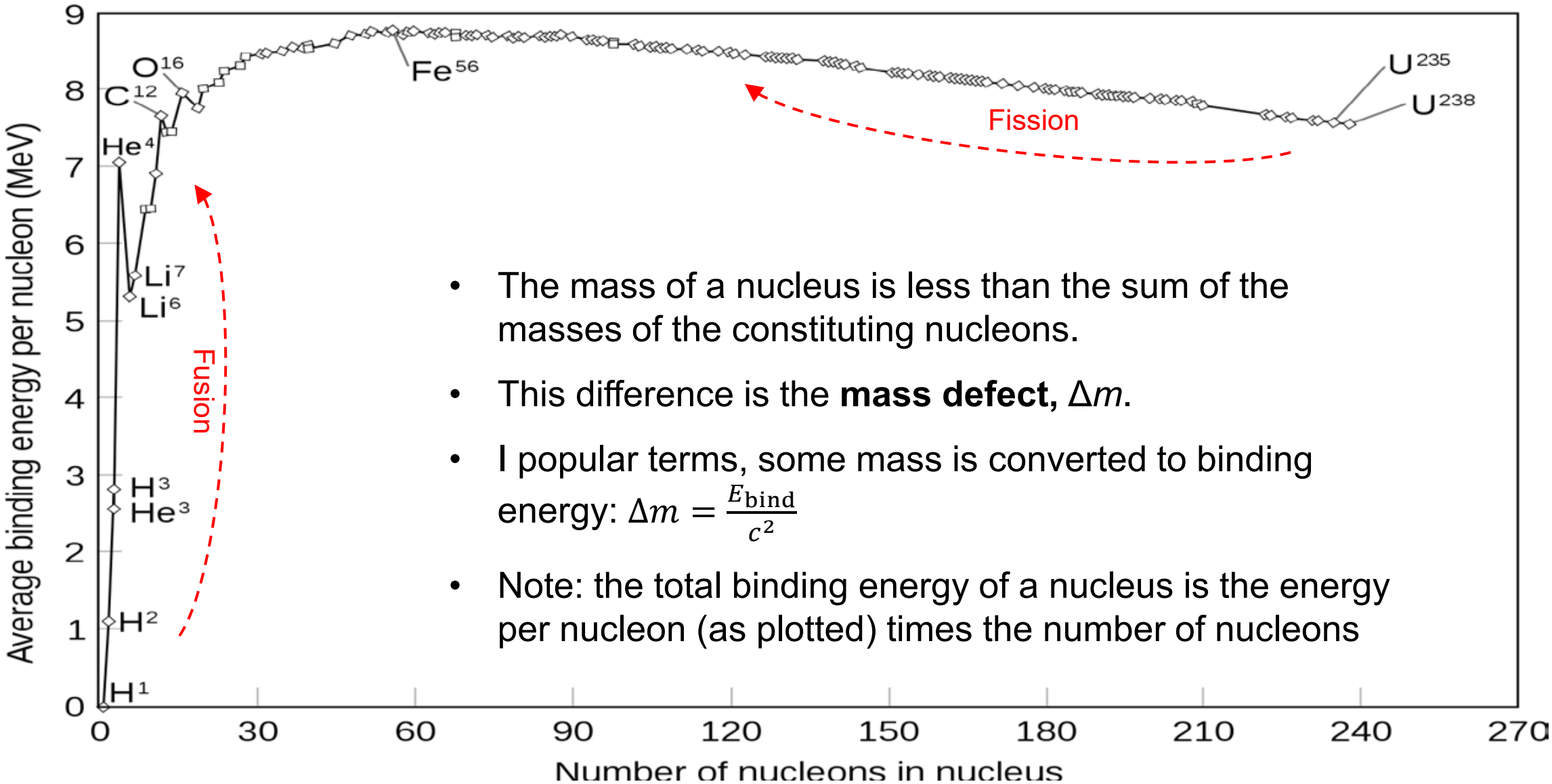


Fission



Fusion

Binding energy and mass defect



- The mass of a nucleus is less than the sum of the masses of the constituting nucleons.
- This difference is the **mass defect,** Δm.
- I popular terms, some mass is converted to binding energy: $\Delta m = \frac{E_{\text{bind}}}{c^2}$
- Note: the total binding energy of a nucleus is the energy per nucleon (as plotted) times the number of nucleons

The energy released by fission or fusion

If the total binding energy of the products is larger than the total binding energy of the reactants, the energy difference is liberated as radiation and kinetic energy of the products.

Definitions:

- **eV** (electronvolt):. A practical energy unit when looking at atoms and smaller particles

  1 eV the numerical charge of an electron (or proton) time 1 V

  $|q_e| = 1.60218 \cdot 10^{-19}$ C, so 1 eV is $1.60218 \cdot 10^{-19}$ J

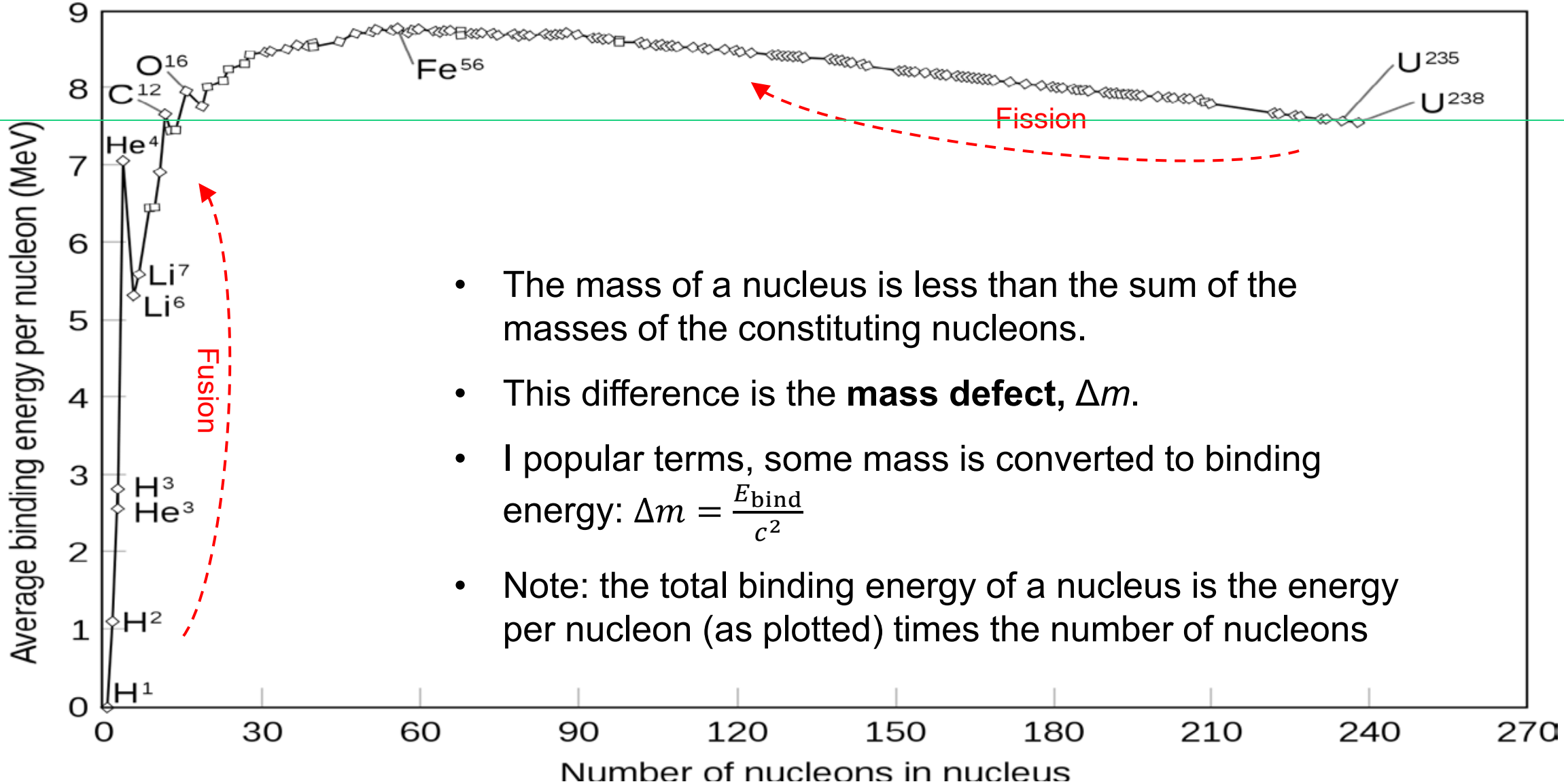
- **amu** (atomic mass unit) or **u** (unit): 1 amu is exactly one twelfth of the mass of a carbon-12 atom ($^{12}_{6}\mathrm{C}$, including the 6 electrons)

  1 amu = 1.6605×10^{-27} kg. 1 amu = 1 dalton (Da). The energy equivalent of 1 amu according is 931.5 MeV

The nuclear binding energy, $^{235}_{92}\mathrm{U}$ as an example

	Number	Mass pr. Particle	Total mass including electrons
Protium ($^{1}_{1}\mathrm{H}$):	92	1.007825 amu ($1.6735 \cdot 10^{-27}$ kg)	92.720 amu ($1.5396 \cdot 10^{-25}$ kg)
Neutron:	143	1.0087 amu ($1.6749 \cdot 10^{-27}$ kg)	144.24 amu ($2.3951 \cdot 10^{-25}$ kg)
Total nucleons:	235		236.96 amu ($3.9347 \cdot 10^{-25}$ kg)
$^{235}_{92}\mathrm{U}$ atom	1	235.04 amu (3.902810^{-25} kg)	235.04 amu ($3.9030 \cdot 10^{-25}$ kg)
Mass defect			1.92 amu ($3.17 \cdot 10^{-27}$ kg)
Binding energy per nucleus (931.5 MeV pr. amu)			1788.5 MeV
Binding energy per nucleon (with 235 nucleons)			7.61 MeV

Binding energy and mass defect



- The mass of a nucleus is less than the sum of the masses of the constituting nucleons.
- This difference is the **mass defect,** Δm.
- I popular terms, some mass is converted to binding energy: $\Delta m = \frac{E_{\text{bind}}}{c^2}$
- Note: the total binding energy of a nucleus is the energy per nucleon (as plotted) times the number of nucleons

Nuclear reactions

Nomenclature and balancing reactions

No. of nucleons (protons + neutrons)

$^{235}_{92}\mathrm{U}$ — Element

No. of protons
= Charge
= Element

Proton: $^{1}_{1}\mathrm{p}$

Neutron: $^{1}_{0}\mathrm{p}$

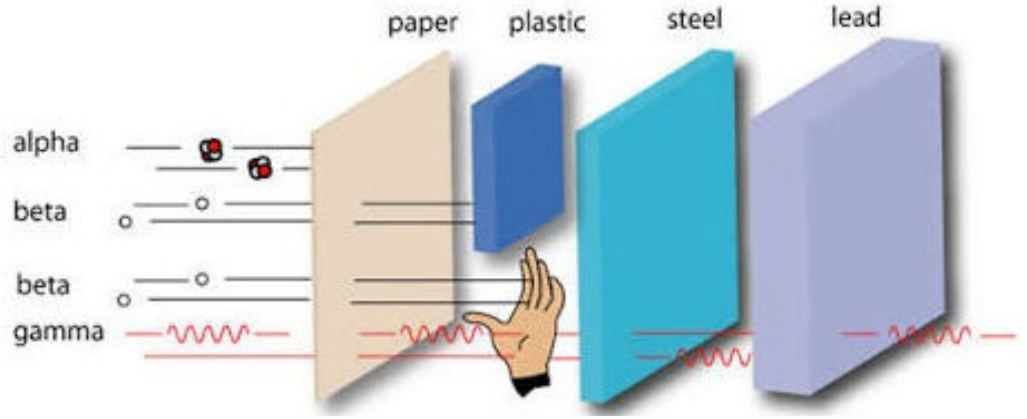
Electron: $^{0}_{-1}\mathrm{e}$

$\Sigma = 236$ $\qquad$ $\Sigma = 236$

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{89}_{36}\mathrm{Kr} + {}^{144}_{56}\mathrm{Ba} + 3\,{}^{1}_{0}\mathrm{n} + \text{energy}$$

$\Sigma = 92$ $\qquad$ $\Sigma = 92$

Radioactivity



Radiation	Nature	Energy	Penetration
Alpha-radiation	α-particles, i.e., helium nuclei, $^{4}_{2}\mathrm{He}$	~ 5 MeV kinetic energy	• Stopped by a few cm of air or a sheet of paper. • Cannot pass the human skin
Beta-radiation	Fast electrons $^{0}_{-1}e$ (or positrons $^{0}_{1}e$, β^- and β^+, respectively)	20 keV to a few MeV.	• Depends on the radiation energy. • Typically, they pass several millimeters into skin or plastic. • With 0.5 MeV the range is about one meter in air.
Gamma-radiation	High energy electromagnetic radiation	From a few keV to about 10 MeV	• Depending on its energy. • Passes though most materials including lead and concrete if the layer is not thick enough.

Radioactivity

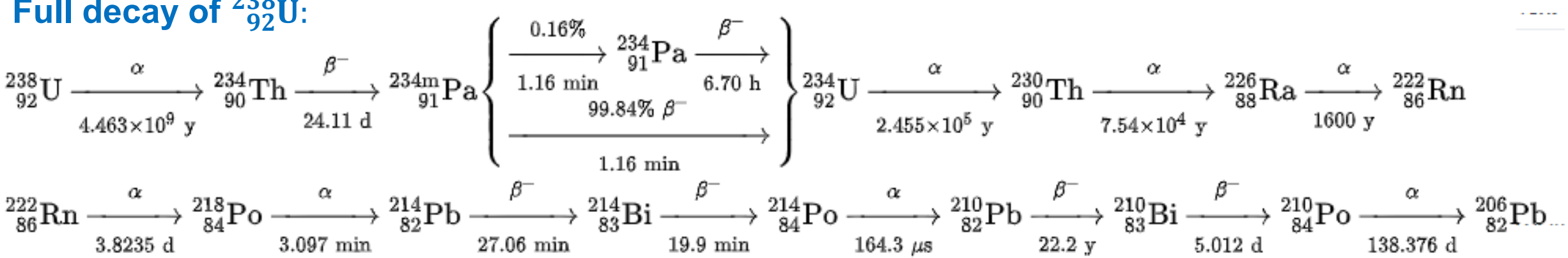
Unstable isotope change to other isotopes spontaneously

Example: $^{238}_{92}\mathrm{U} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{234}_{90}\mathrm{Th} + \mathrm{energy}$

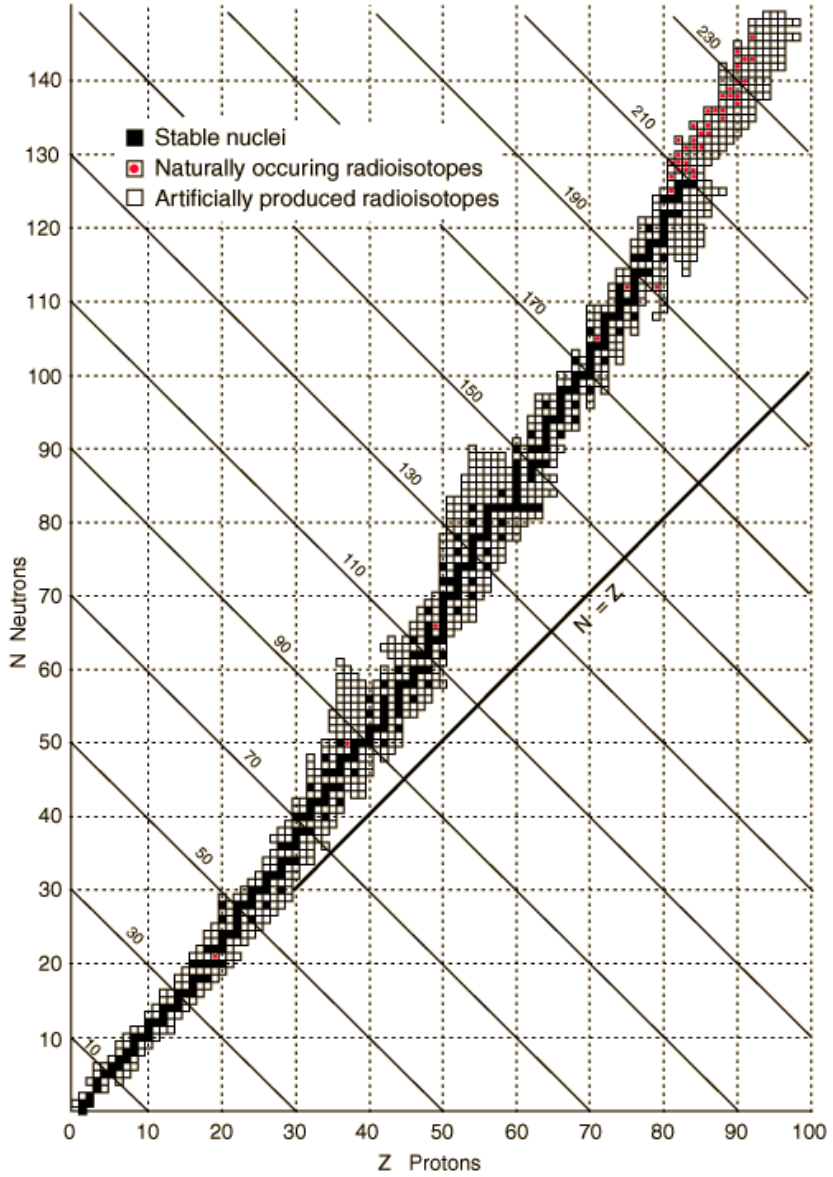
- Process accompanied by radiation
- Number of nucleons constant
- Mass *not* constant
- Unstable isotopes are assigned a half life (nano-seconds to billions of years)

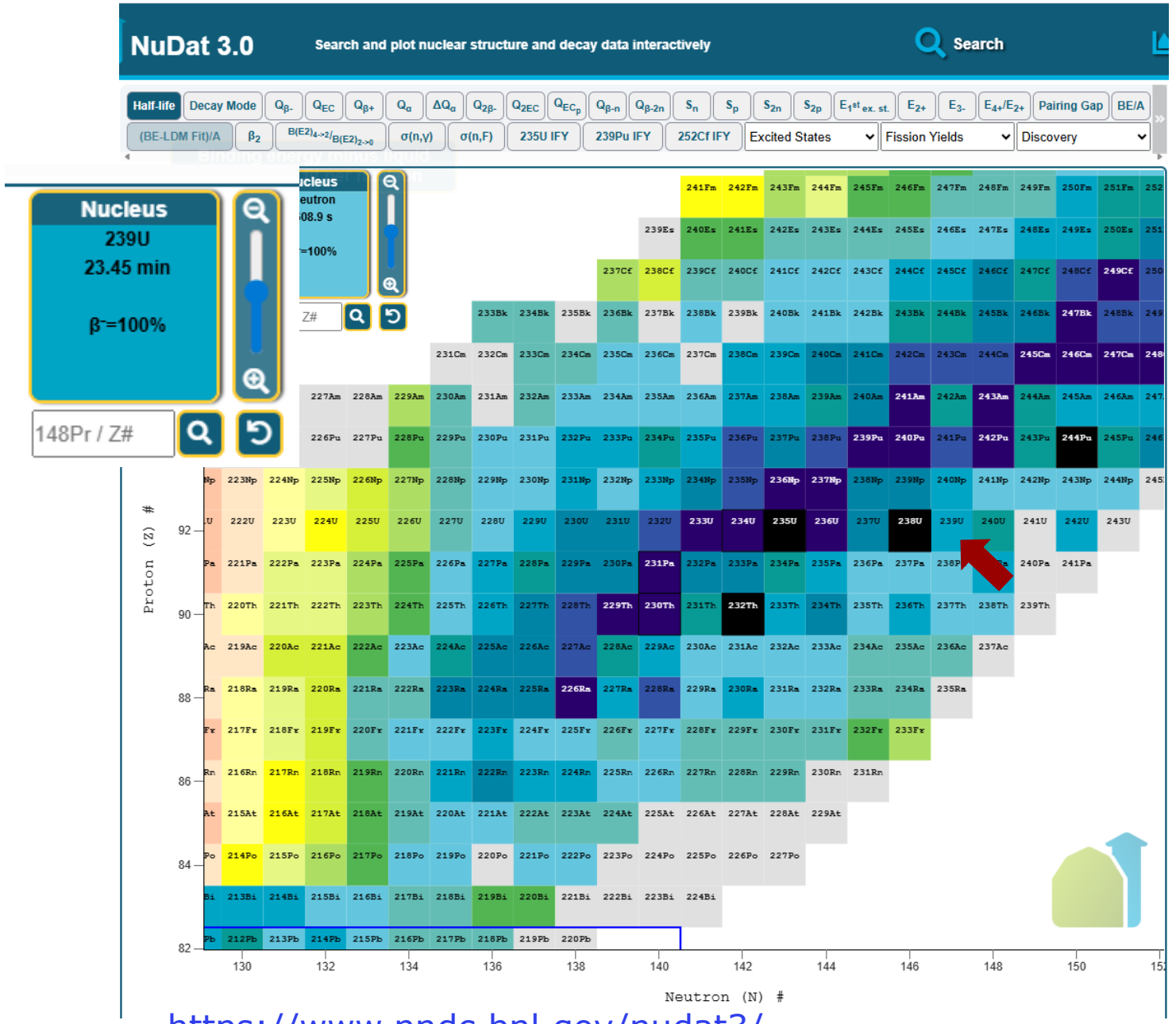
$$^{222}_{92}\mathrm{Np}: 380\ ns, \qquad ^{238}_{92}\mathrm{U}: 4.5{\cdot}10^{9}\ \mathrm{y}$$

Full decay of $^{238}_{92}\mathrm{U}$:



Isotope charts





https://www.nndc.bnl.gov/nudat3/

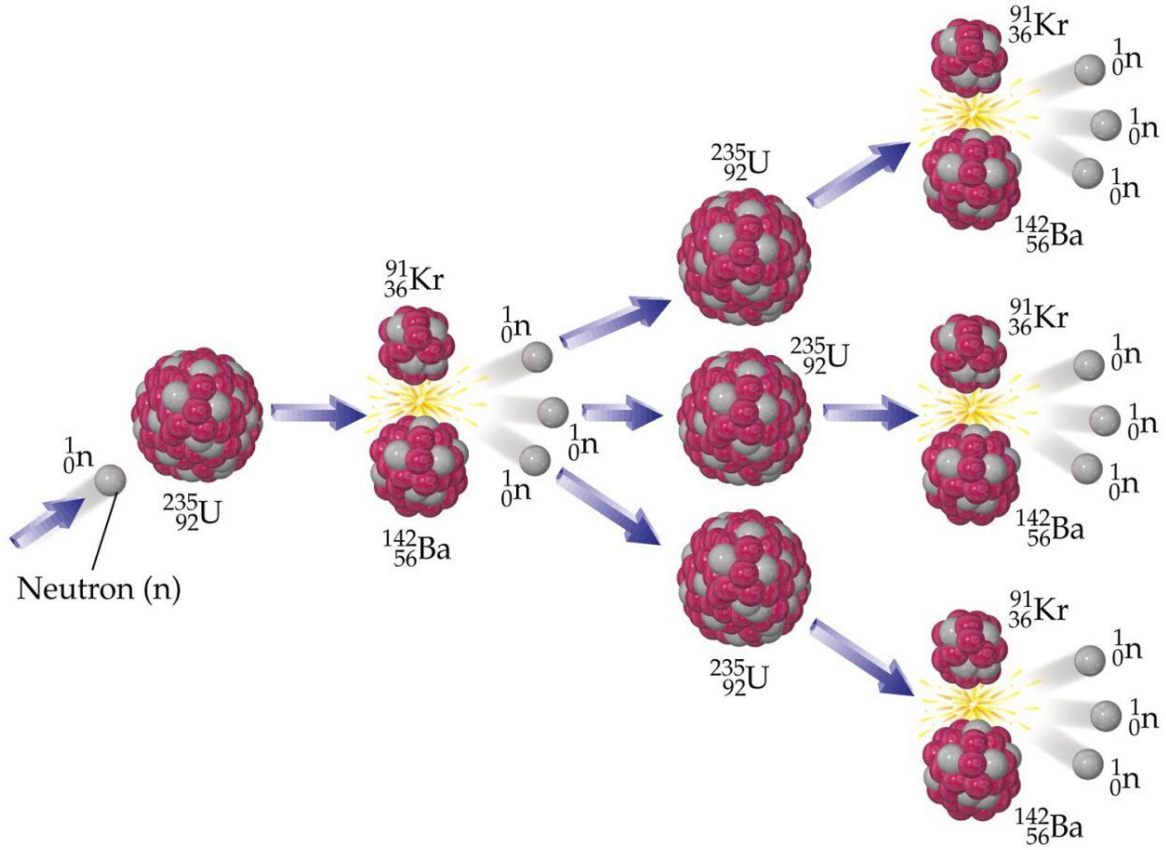
Fissile isotopes

Only thorium (Th), uranium (U) and plutonium (Pu) are practical for fission

Need to liberate multiple neutrons during fission

Directly fissile isotopes

- Only four directly fissile isotopes are known: $^{233}_{92}\mathrm{U}$, $^{235}_{92}\mathrm{U}$, $^{239}_{94}\mathrm{Pu}$ and $^{241}_{94}\mathrm{Pu}$
- Only $^{235}_{92}\mathrm{U}$ exists in nature (0.7 % of natural uranium. The rest is $^{238}_{92}\mathrm{U}$)
- Only slow neutrons are effective in fission (named thermal neutrons)
- Neutrons liberated from chain reaction are fast – must be slowed down by moderator



Fertile isotopes

Isotopes that can be converted to one of the four direct fissile isotopes by absorption of fast neutrons.

Most common are $^{232}_{90}\mathrm{Th}$, $^{238}_{92}\mathrm{U}$

$$^{232}_{90}\mathrm{Th} + ^{1}_{0}\mathrm{n} \rightarrow {}^{233}_{90}\mathrm{Th} \xrightarrow{\beta^-} {}^{233}_{91}\mathrm{Pa} \xrightarrow{\beta^-} \mathbf{^{233}_{92}U}$$

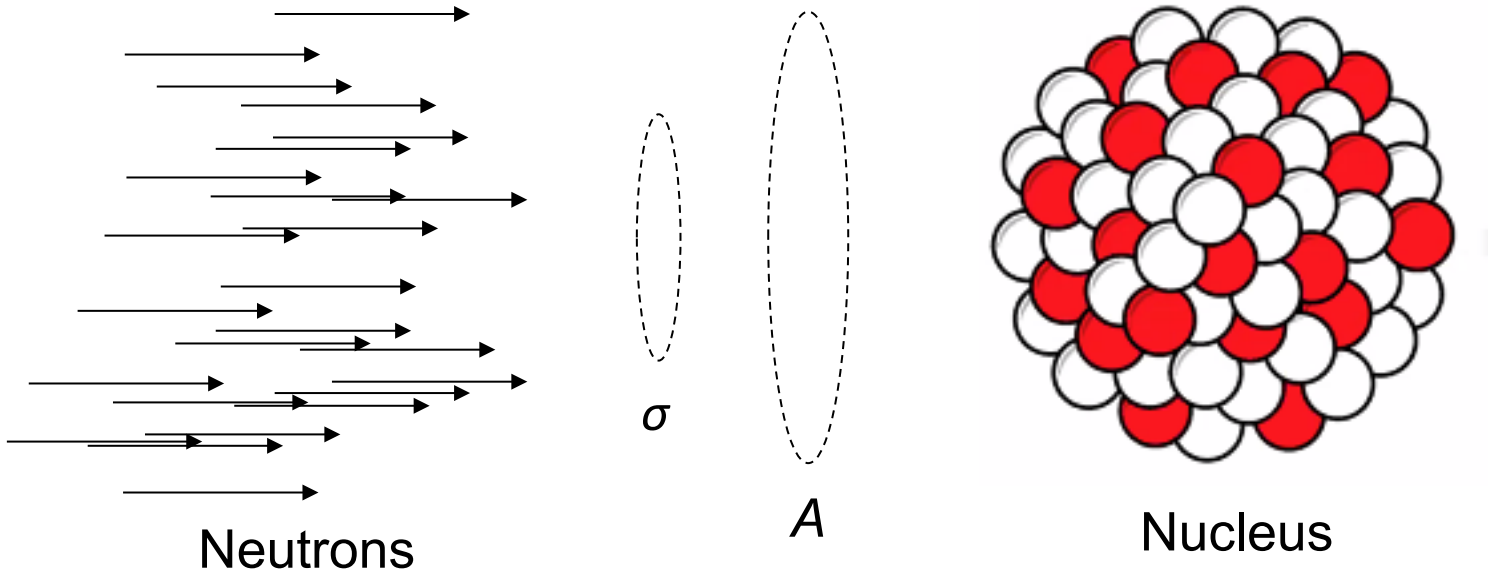
$$^{238}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow {}^{239}_{92}\mathrm{U} \xrightarrow{\beta^-} {}^{239}_{93}\mathrm{Np} + \xrightarrow{\beta^-} \mathbf{^{239}_{94}Pu}$$

Fertile isotopes are long-lived isotopes that are not fissile themselves

Neutron cross section for fission or absorption

Neutrons only cause fission sometime.

The probability that a neutron will induce fission, is described by the cross section, σ, for fission



All neutrons passing through *A* have a possibility of being absorbs and causing fission.

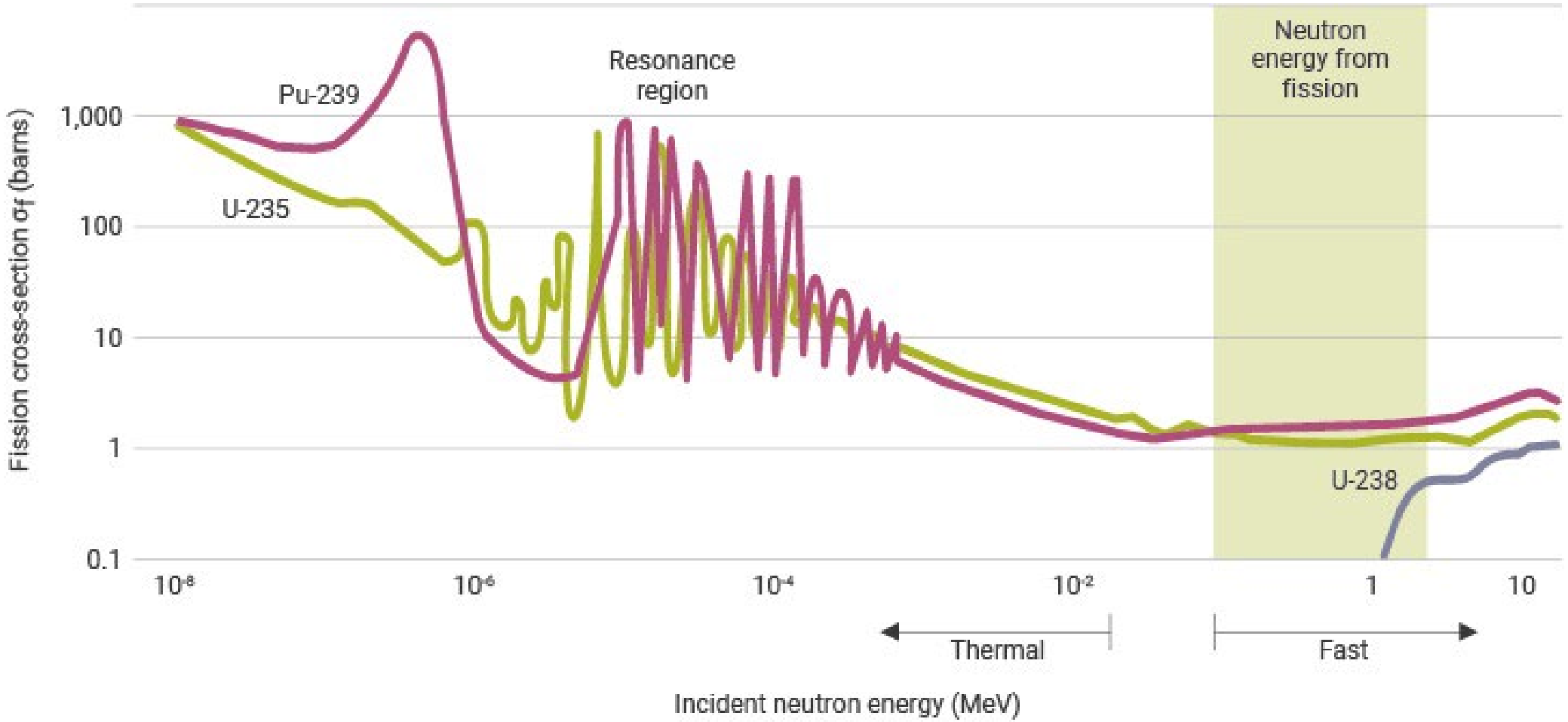
The successful fraction of the neutrons can be assigned to a fraction of the area, σ

σ is given in the unit barn.

1 barn = 10^{-28} m^2.

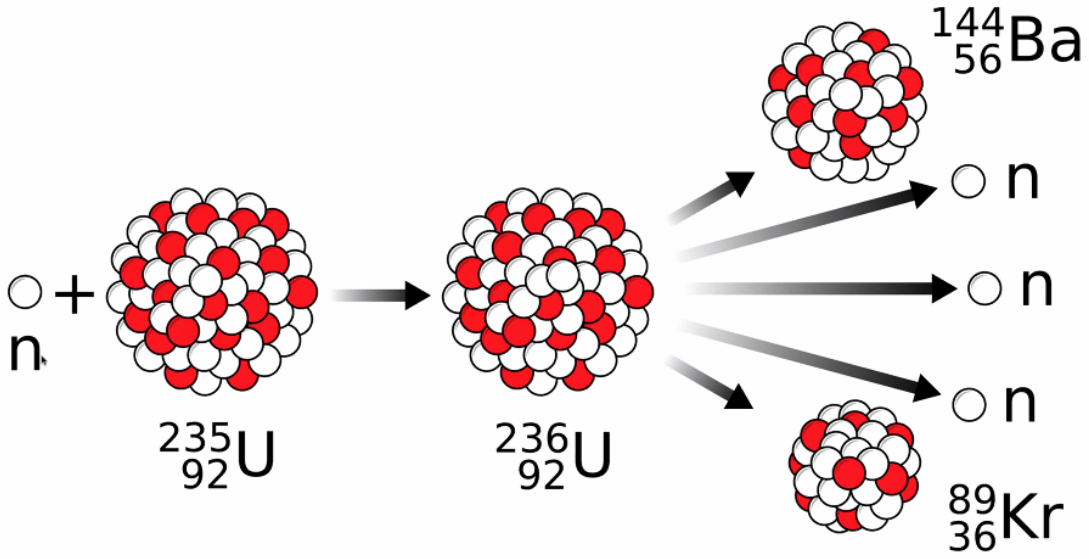
A nucleus is around 10^{-24} cm^2 = 10^4 barn

Fission cross section



https://world-nuclear.org/information-library/nuclear-fuel-cycle/introduction/physics-of-nuclear-energy

Fission products (many possibilities)



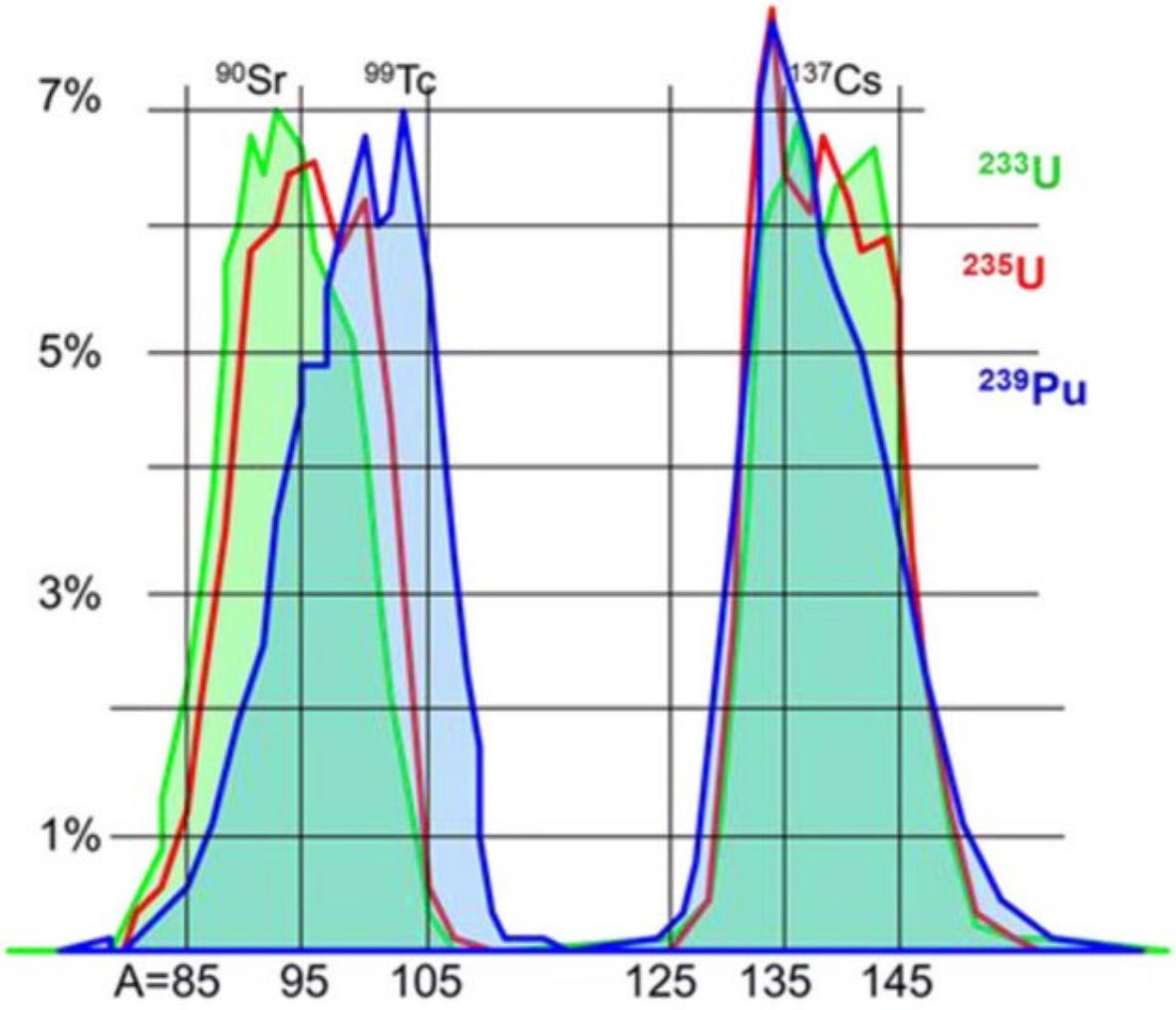
$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow ^{100}_{38}\mathrm{Sr} + ^{134}_{54}\mathrm{Xe} + 2\,^{1}_{0}\mathrm{n} + \text{energy}$$

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow ^{89}_{36}\mathrm{Kr} + ^{144}_{56}\mathrm{Ba} + 3\,^{1}_{0}\mathrm{n} + \text{energy}$$

General:

$$^{235}_{92}\mathrm{U} + ^{1}_{0}\mathrm{n} \rightarrow ^{236}_{92}\mathrm{U} \rightarrow \mathrm{N1} + \mathrm{N2} + \mathrm{x}\,^{1}_{0}\mathrm{n} + \text{energy}$$

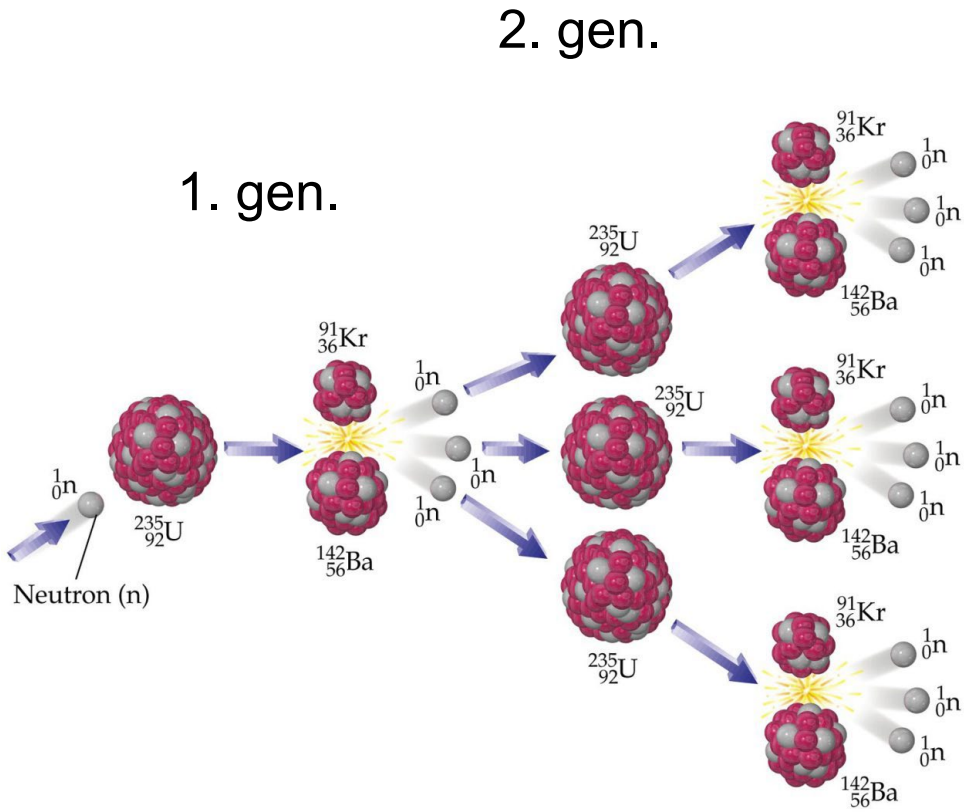
Fission products and energy



Average Energy Release per Fission (MeV)

	^{235}U	^{239}Pu	^{233}U
Prompt Energy			
Kinetic energy of fission fragments	169	176	168
Radiation of prompt γ-rays	7	8	8
Kinetic energy of prompt neutrons	5	6	5
Energy of nonfission neutrons	9	12	9
Delayed (Decay) Energy			
Energy of β-decay	7	5	5
Radiation of γ-rays from decay	9	7	7
Energy of antineutrinos	6	5	5
Total	212	219	207

Criticality



Multiplication factor:

$$k_{\text{eff}} = \frac{\text{no. of fissions, generation } i+1}{\text{no. of fissions, generation i}}$$

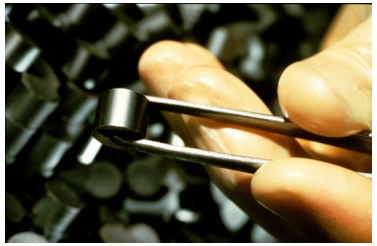
k_{eff} = 1: system critical (chain reaction just sustained)

k_{eff} < 1: system sub-critical (reaction dying out)

k_{eff} > 1: system supercritical (risk of runaway)

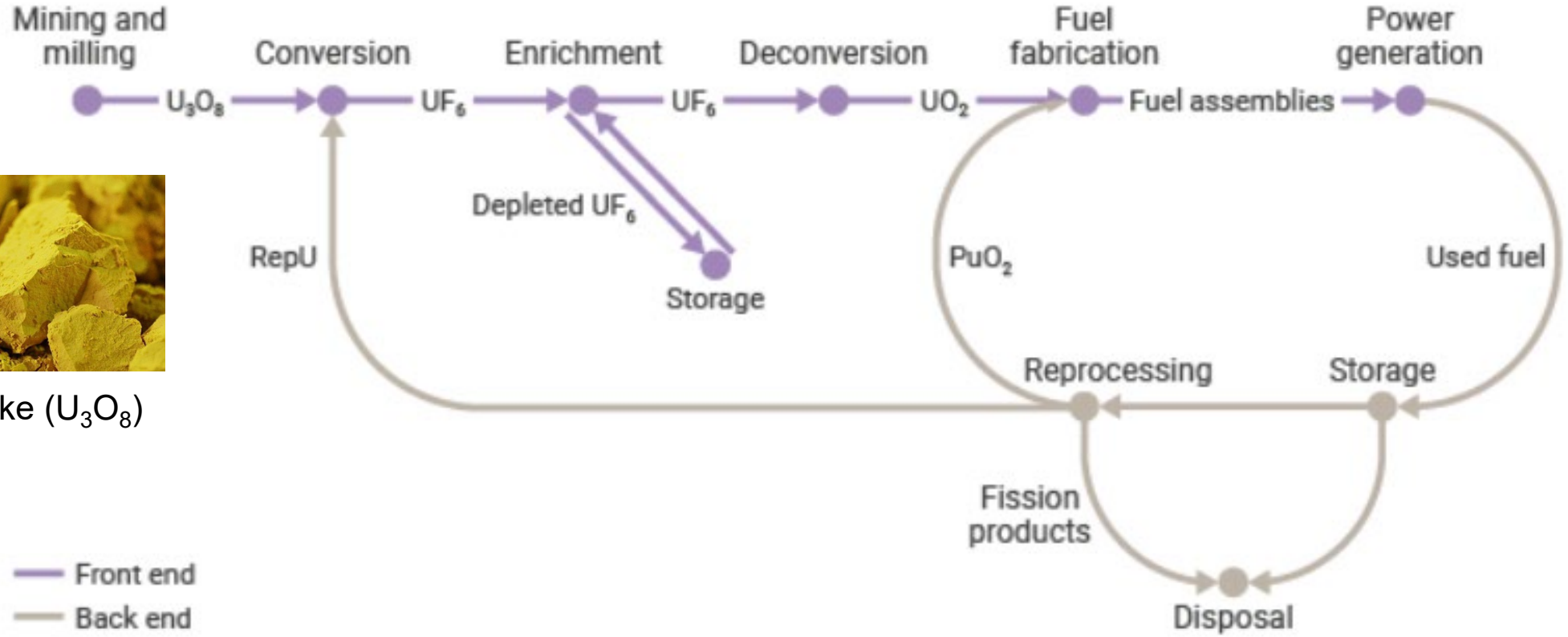
Utilizing nuclear energy

The uranium fuel cycle





Yellow cake (U_3O_8)

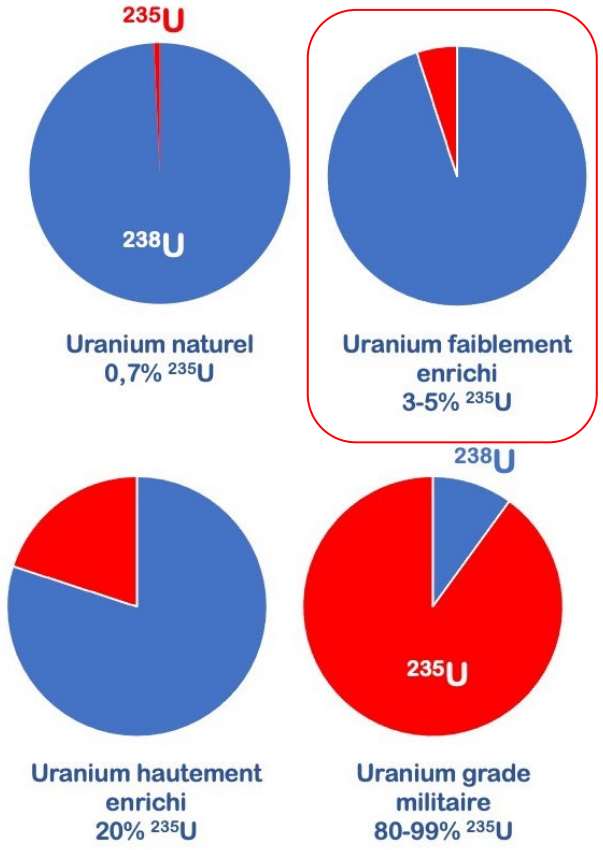


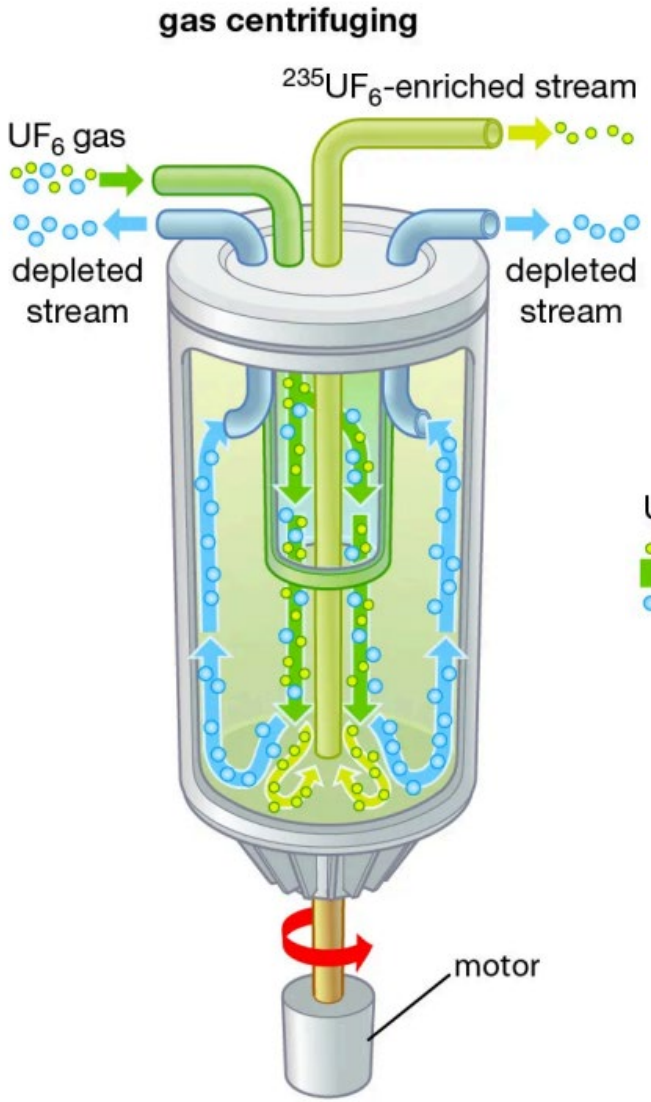
Source: World Nuclear Association

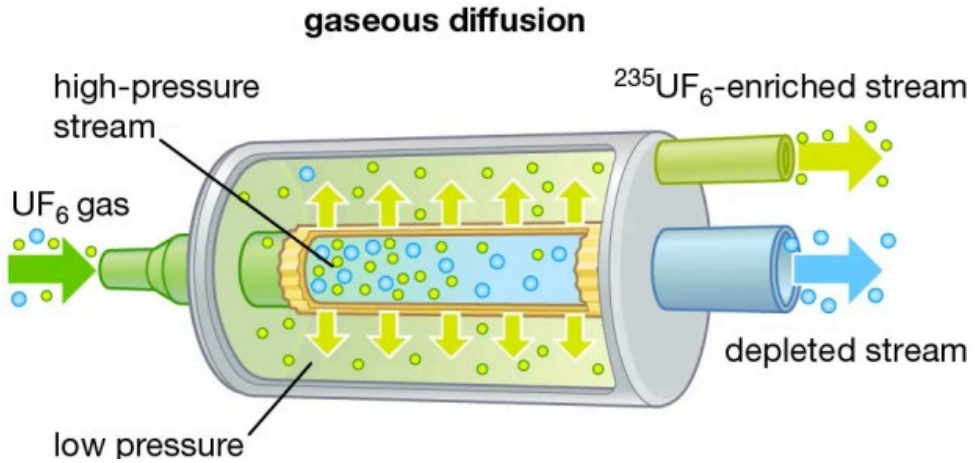
Enrichment – separation of UF_6

Mined:

0.7% ^{235}U, 99.3% ^{238}U





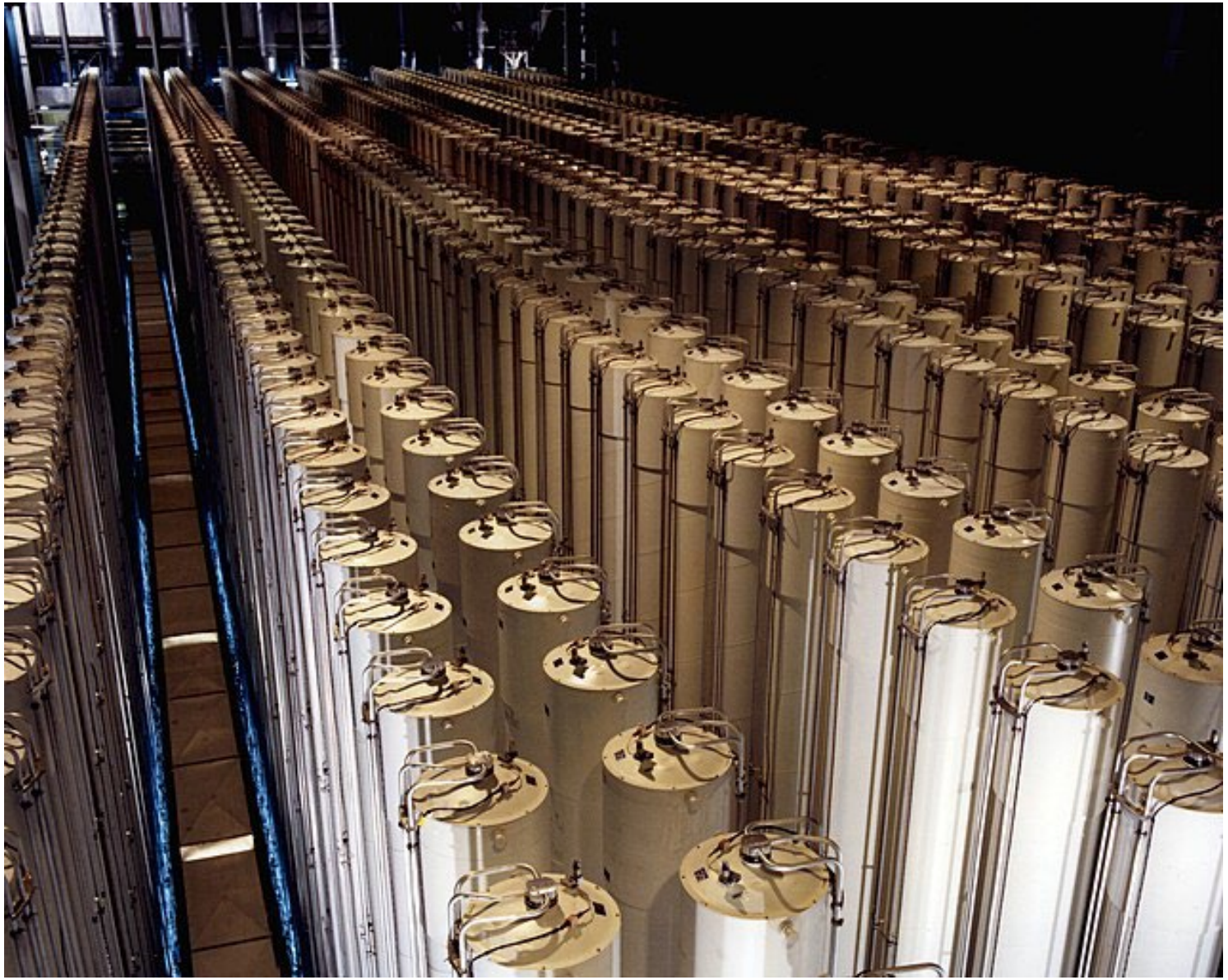


Enrichment

Cascade of gas centrifuges enrichment of uranium.

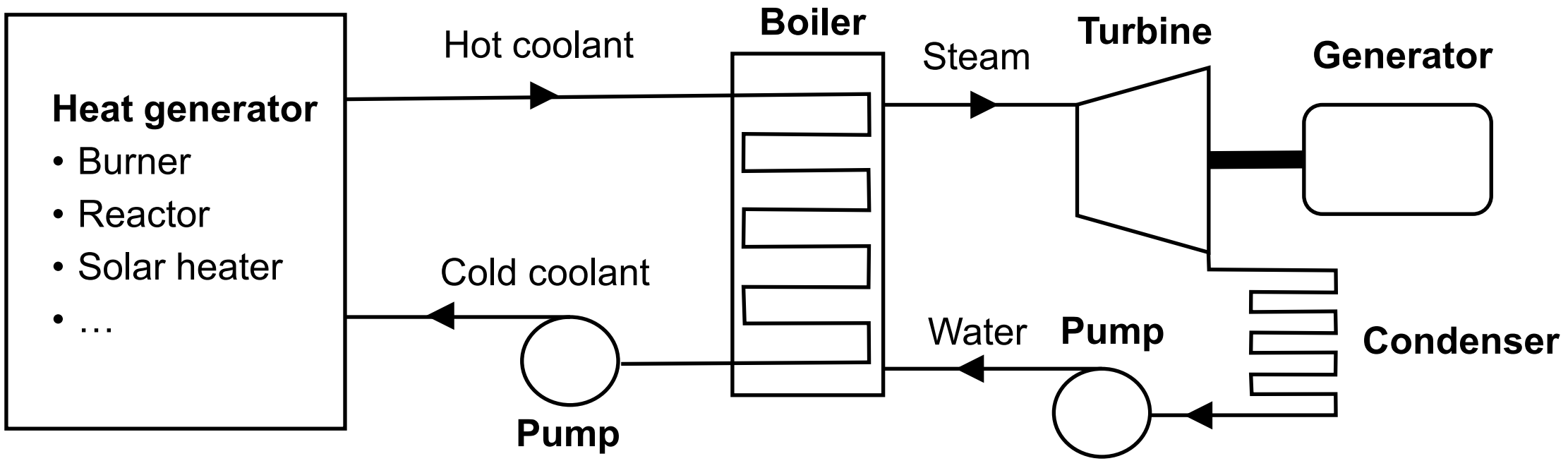
Each centrifuge is 12 m tall.

Piketon, Ohio, 1984.

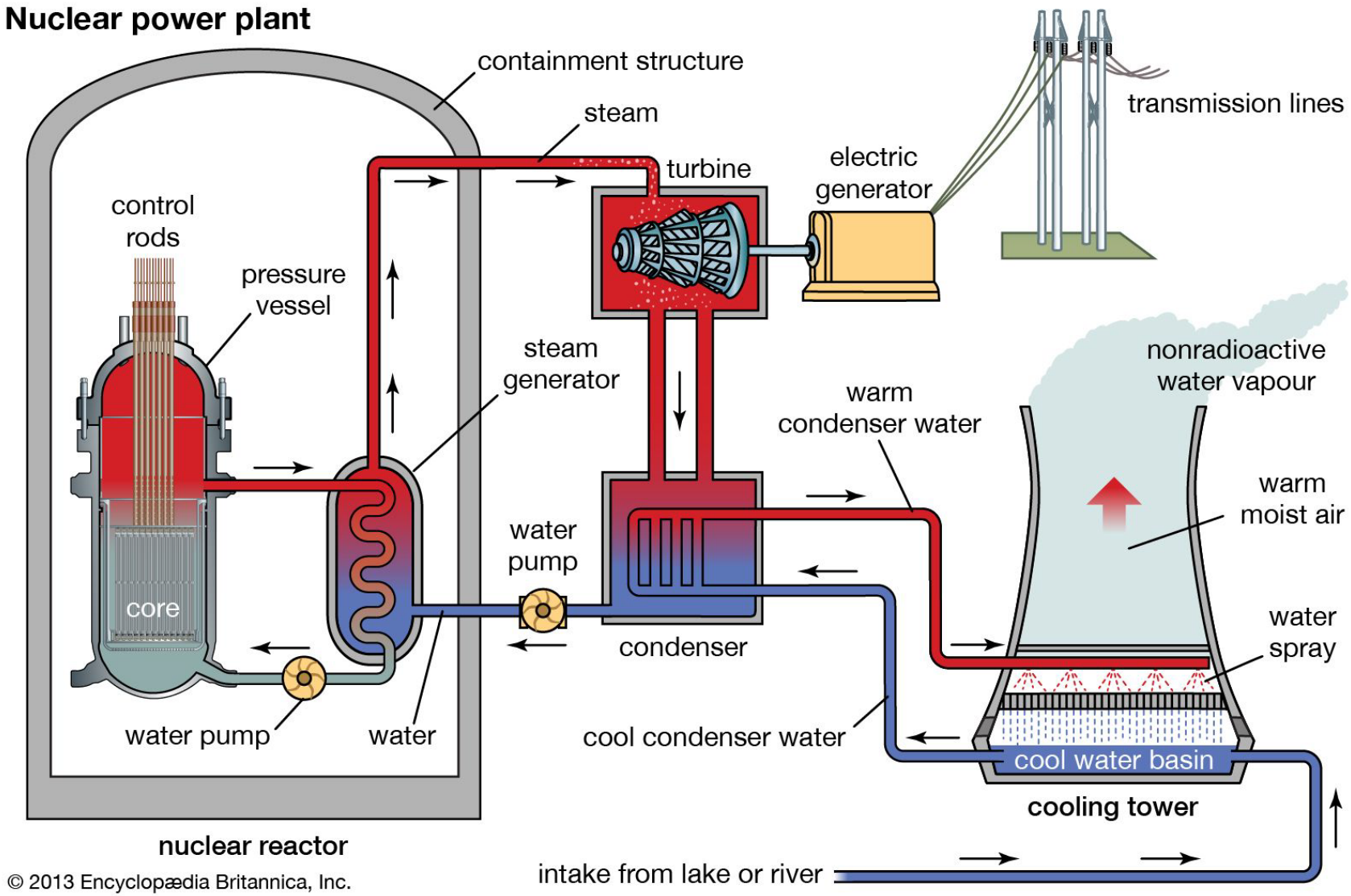


Nuclear reactors

General principle in <u>any</u> thermal power plant



The nuclear power plant





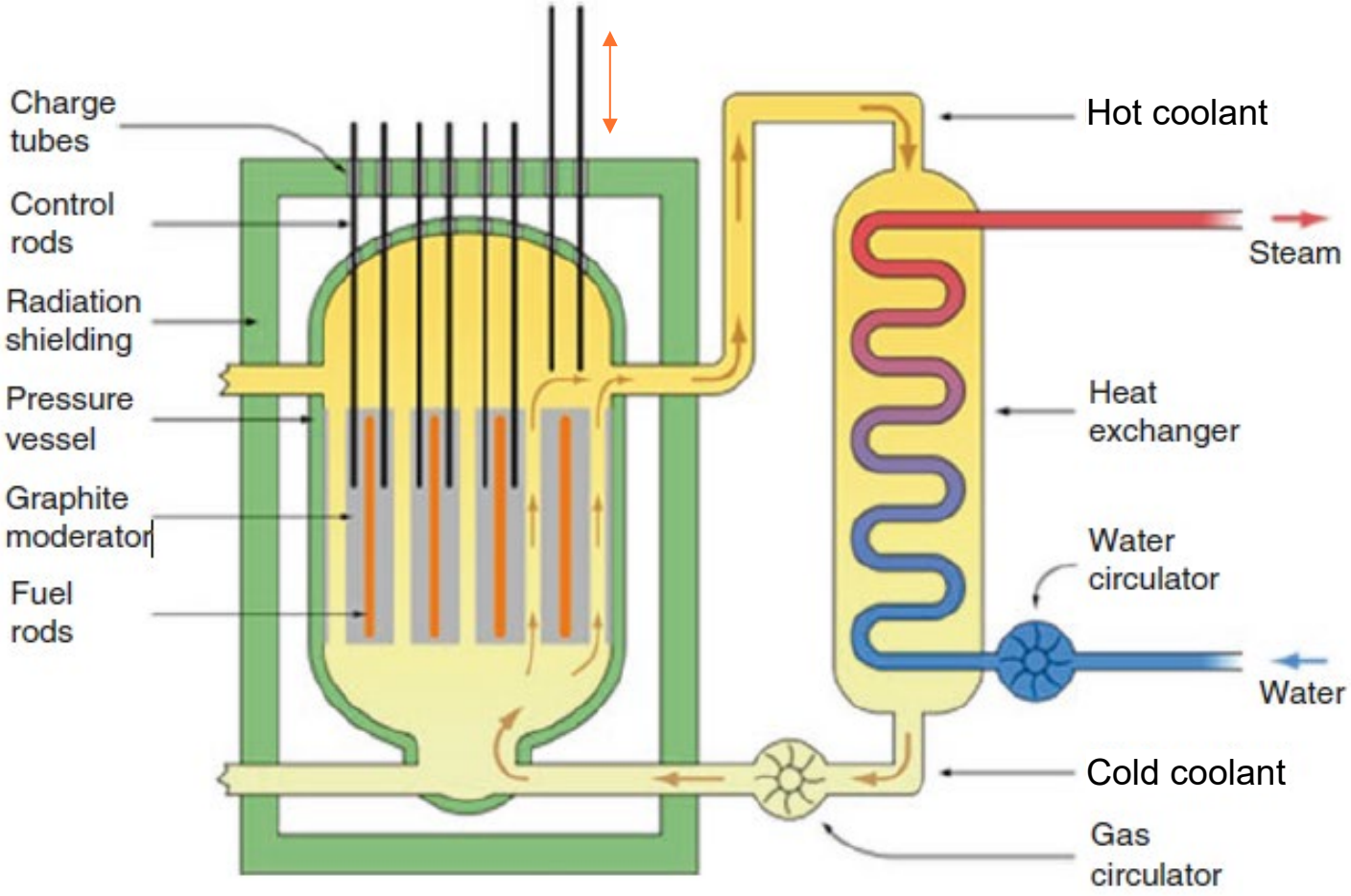
Components of a nuclear fission reactor

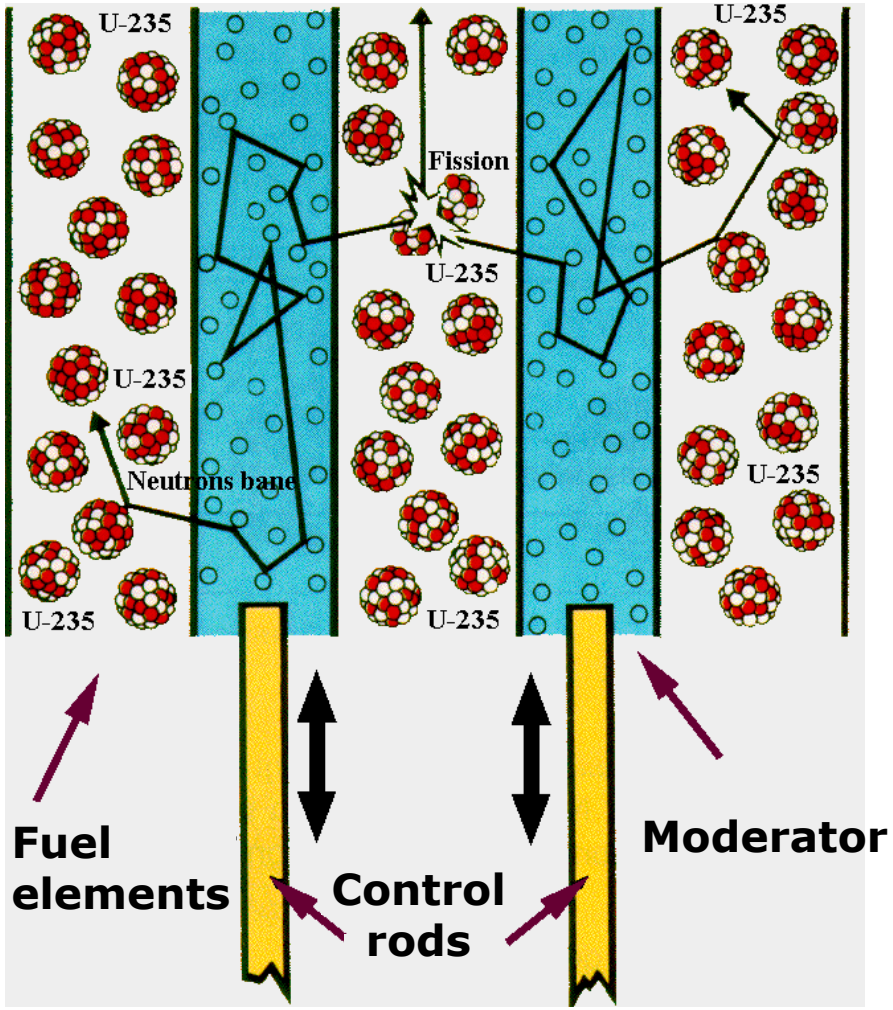
- **Nuclear fuel**
  Uranium, Plutonium, Thorium
- **Moderator**
  Medium to slow down the neutrons from fast neutrons to thermal neutrons
  Water, graphite or heavy water (D_2O)
- **Primary Coolant**
  Water, steam, gas or liquid metal
- **Control rods**
  Medium to control the process (just maintaining criticality) by absorbing the neutrons
  Bor (B), Cadmium (Cd) or Hafnium (Hf)

Moderators

Moderator	Advantages	Disadvantages
Light water	• Good moderator • Inexpensive	• Neutron absorber • Tritium production
Heavy water	• No absorption • Natural uranium	• Expensive • Tritium production
Graphite	• No absorption • Inexpensive	• Poor moderation • C-14 waste

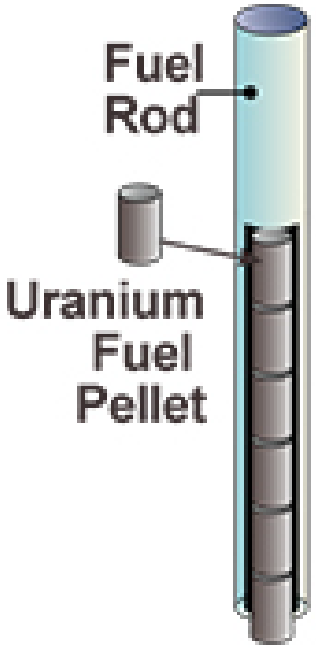
The reactor

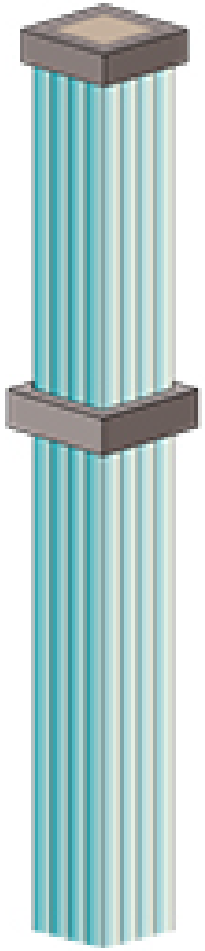


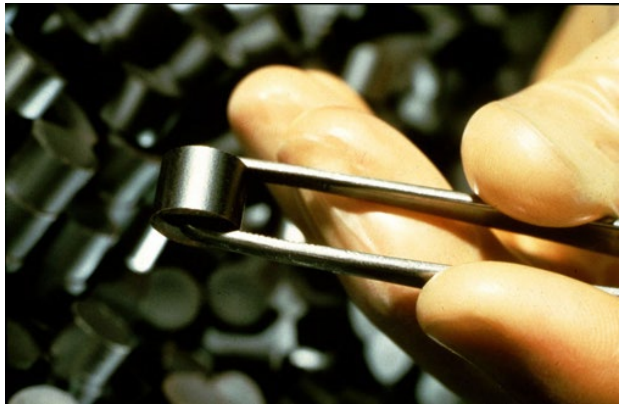


Drawing: Osterreichisches Ökologie-Institut

Fuel elements







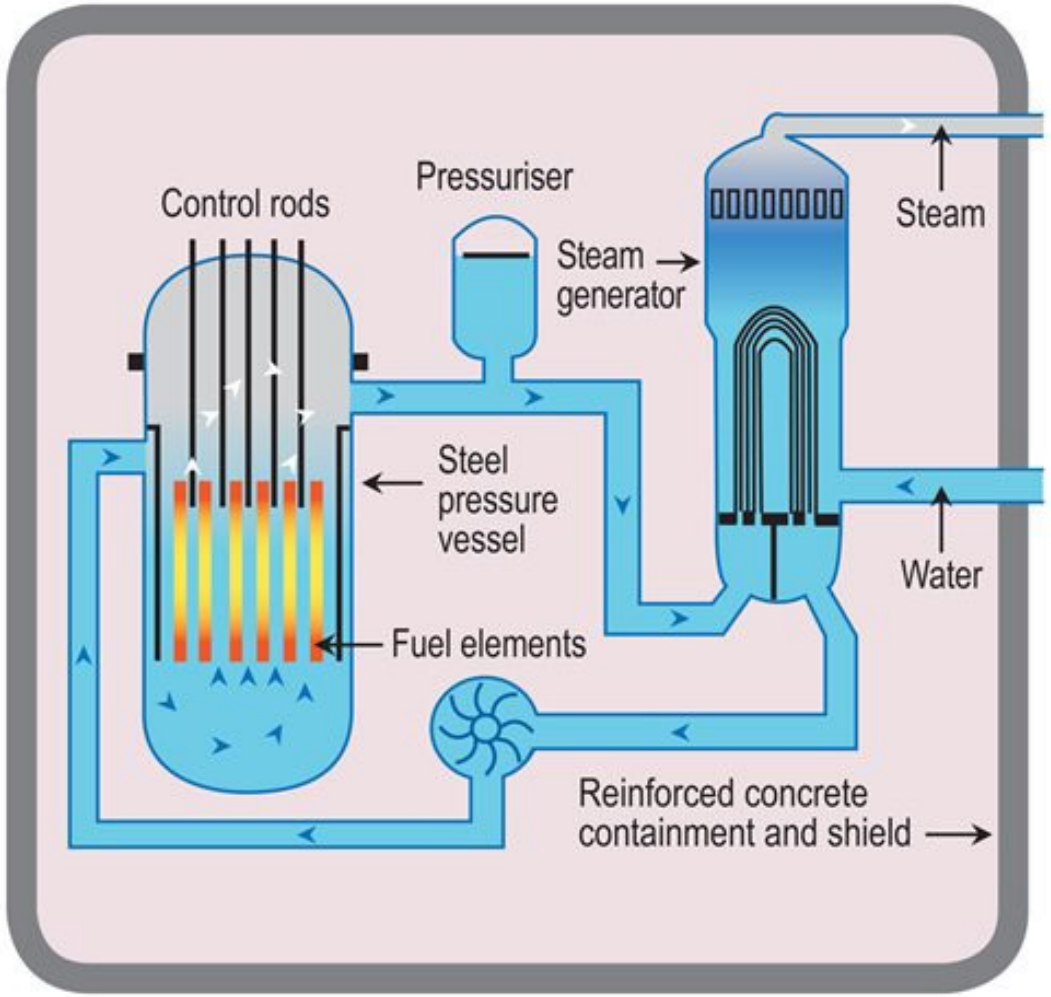
Fuel assembly

Spent fuel assemblies are typically 14 feet (4.3 meters) long and contain nearly 200 fuel rods for PWRs and 80–100 fuel rods for BWRs.

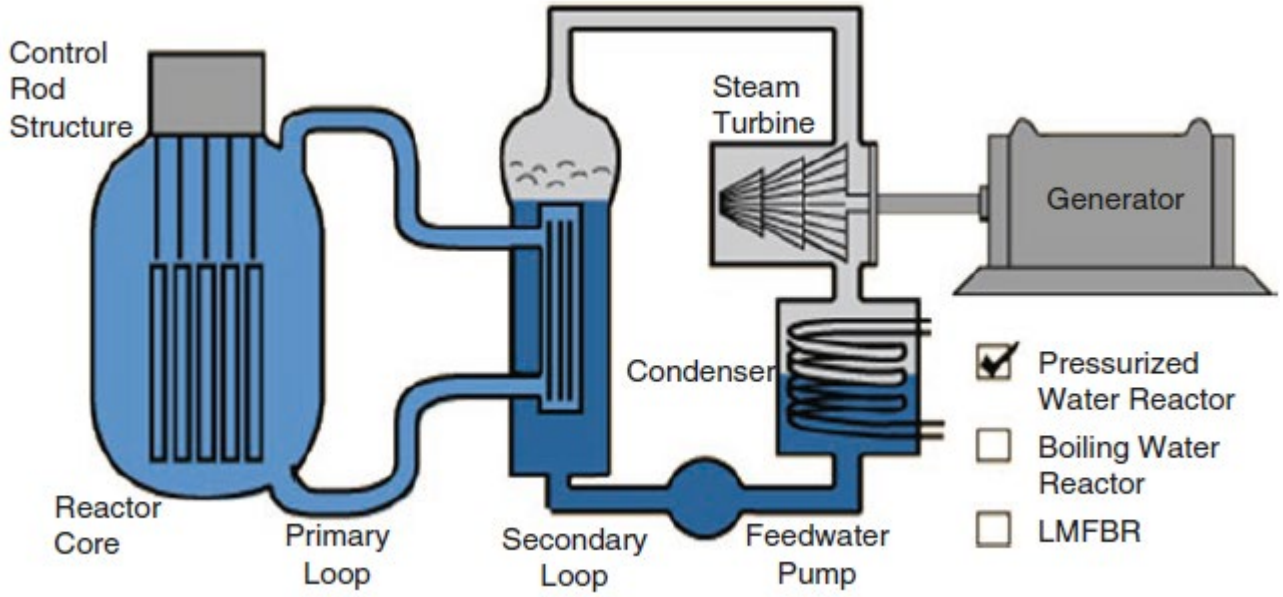


Main types of nuclear ractors today

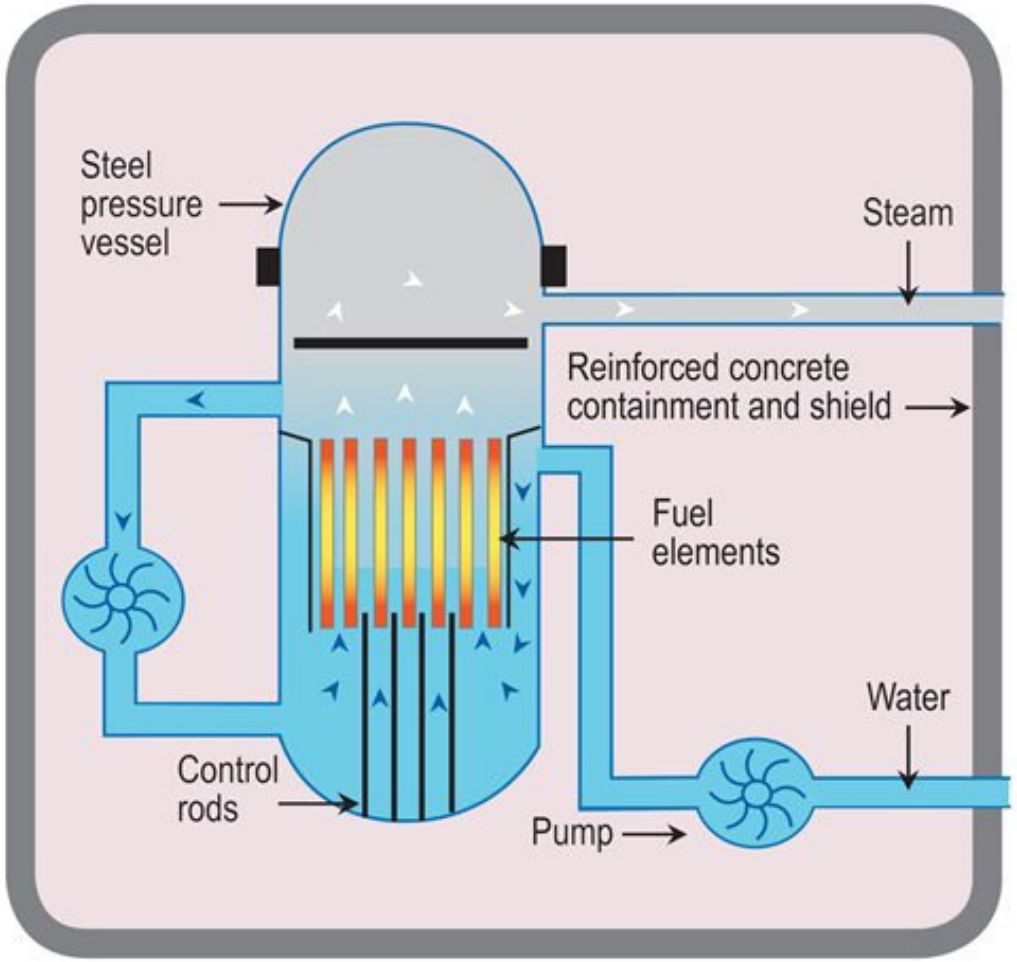
Pressurized water reactor (PWR)



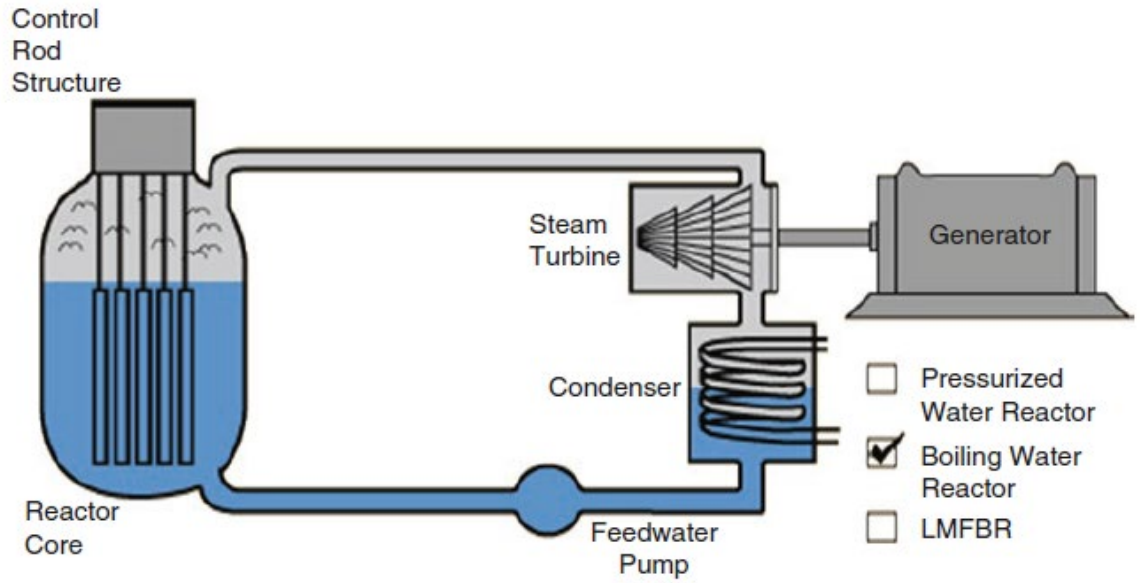
- **A light-water reactor (LWR)**
- About 70% of the global reactor fleet.
- Light water as primary coolant as well as moderator
- Primary cooling unit of pressurized liquid water (ca. 150 bar)
- Water in the secondary circuit boil under low pressure
- Steam out at ca. 325 °C



Boiling water reactor (BWR)



- **A light-water reactor (LWR)**
- The second most common reactor type globally,
- Ca. 15% of the global fleet.
- Light water as primary coolant as well as moderator
- Single coolant loop. Steam directly to the turbine
- BWRs are primarily found in the US, Japan, Sweden and Taiwan.
- Steam out at ca. 290 °C



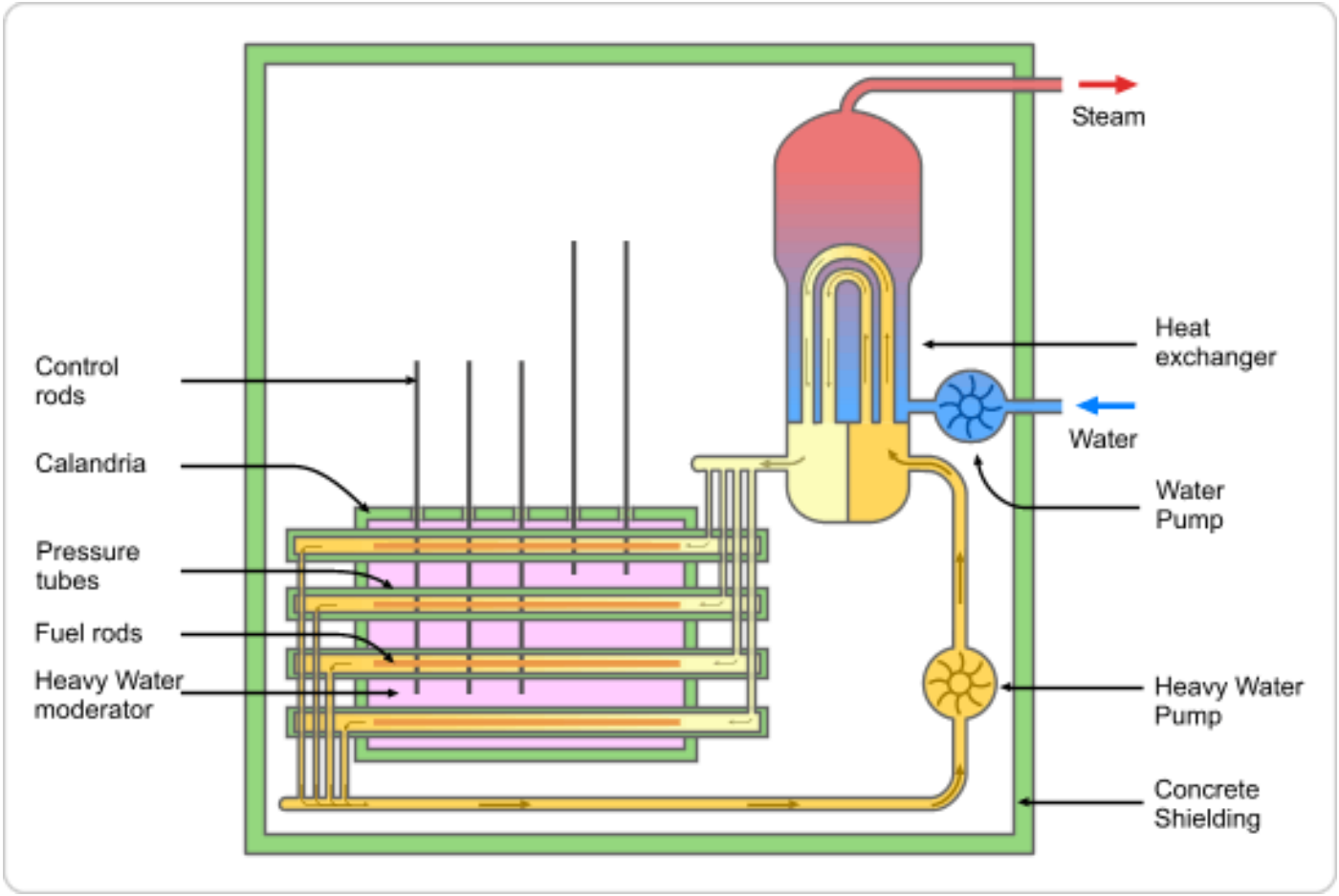
Pressurized heavy-water reactor (PHWR)

Heavy water is a more efficient moderator.

Does not absorb neutrons (light water does to some extent)

The heavy water reactor can run on unrefined uranium.

Canadian technology: CANDU (Canada deuterium uranium reactor).



Main types of nuclear power plants

PWR: Pressurized water reactor

BWR: Boiling water reactor

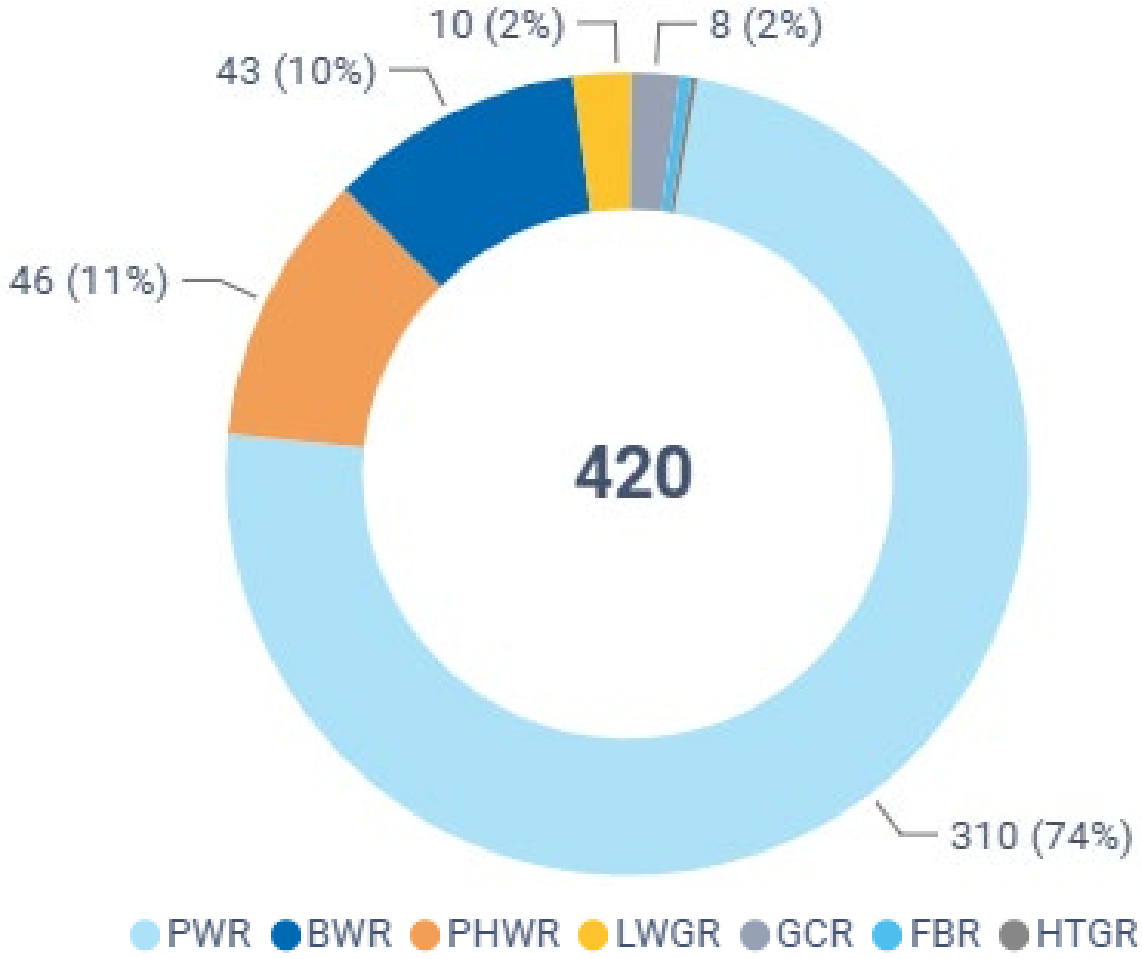
PHWR: Pressurized heavy water reactor

LWGR: Light-water gas cooled reactor

GCR: Gas cooled reactor

FBR: Fast breeder reactor

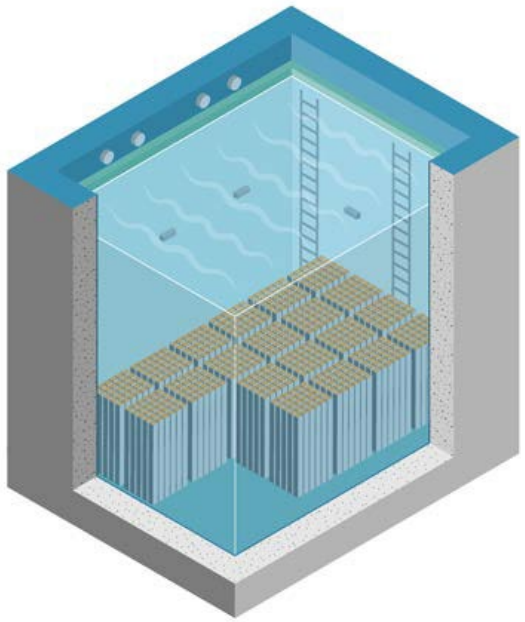
HTGR: High temperature gas-cooled reactor

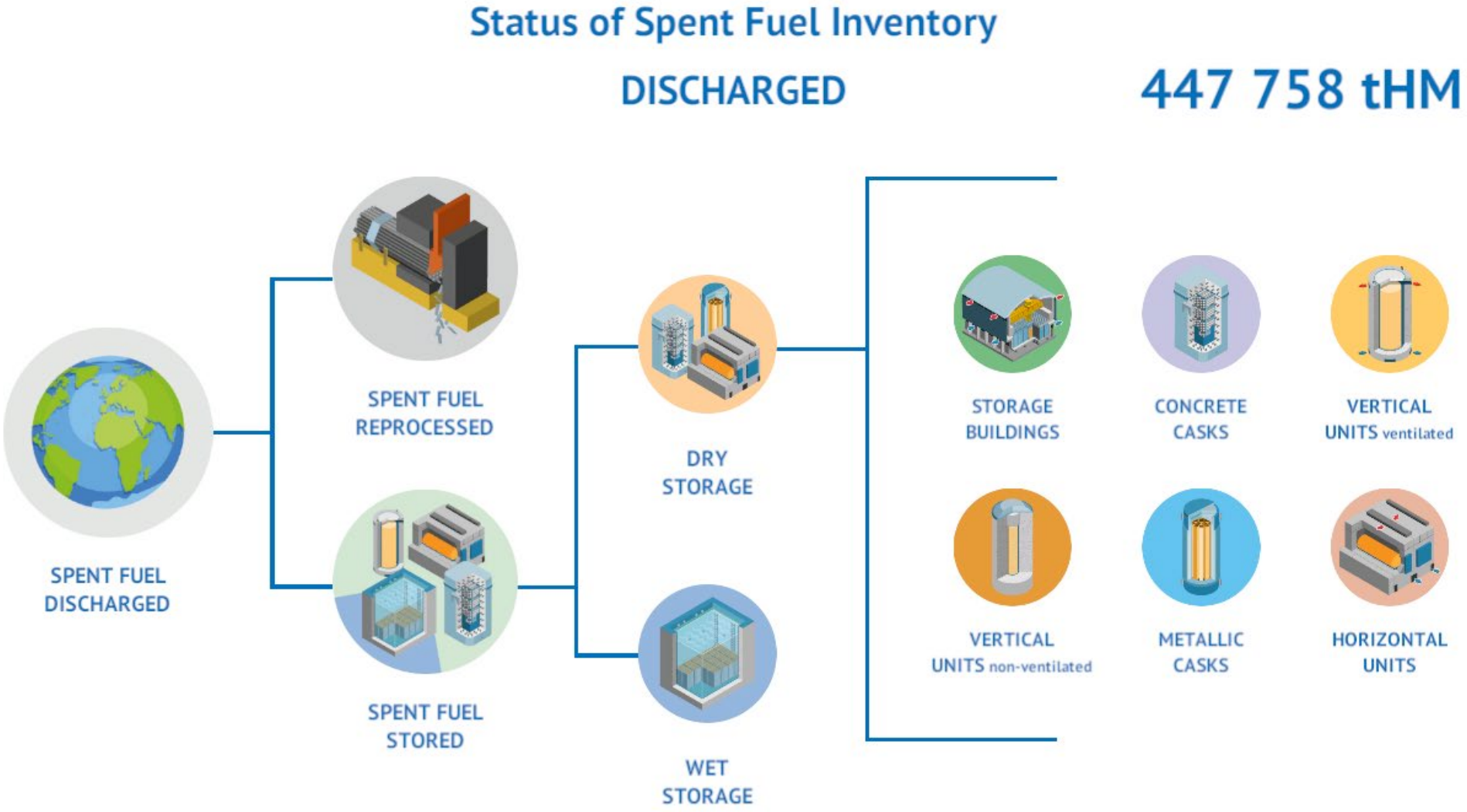


Source: IAEA, PRIS, Power Reactor Information system, 2026

Waste management

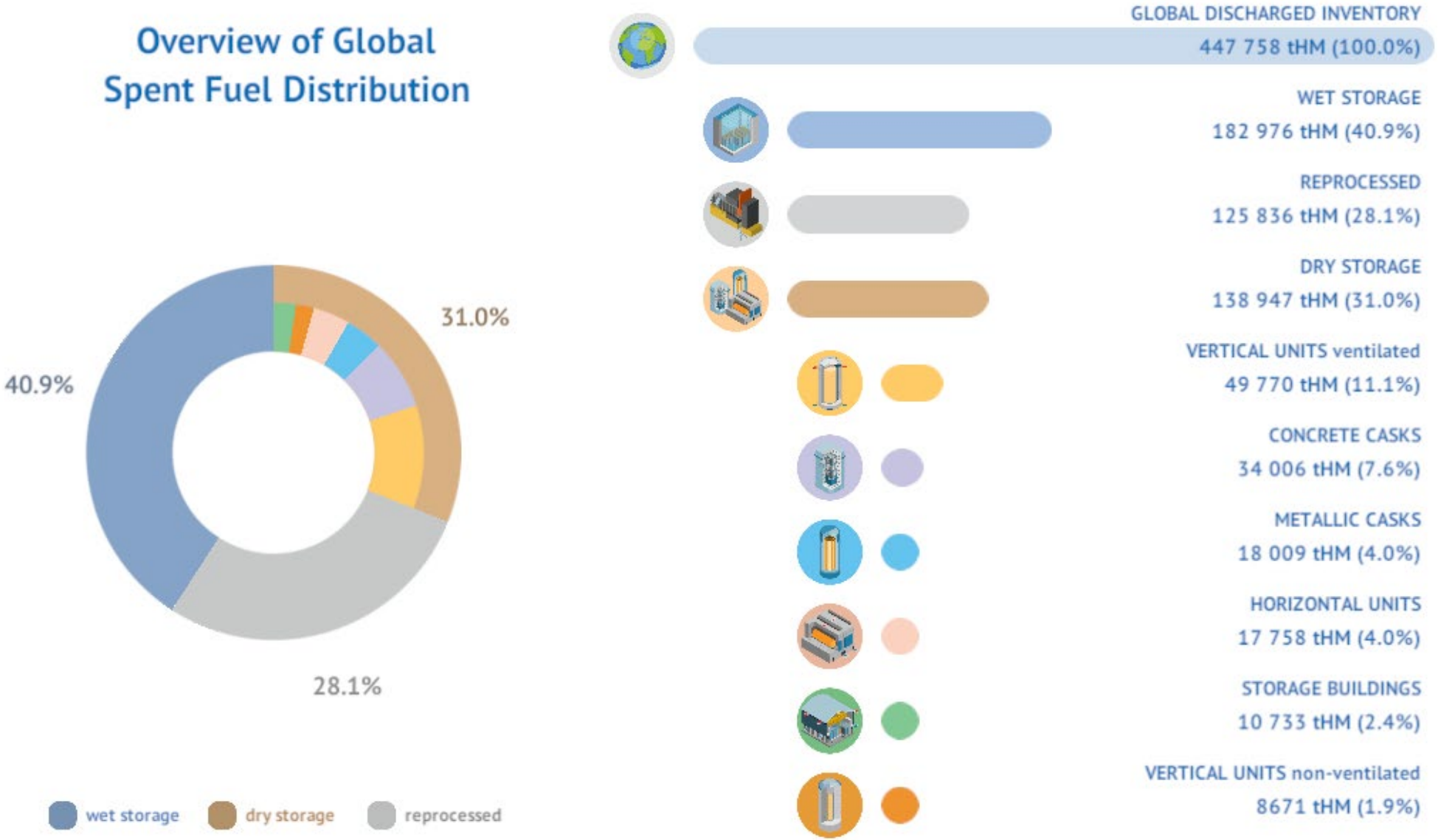
Storage of spent fuel





https://nucleus.iaea.org/sites/connect/SFMpublic/SitePages/SNF-Inventory-status.aspx

Spent nuclear fuel (SNF)



https://nucleus.iaea.org/sites/connect/SFMpublic/SitePages/SNF-Inventory-status.aspx

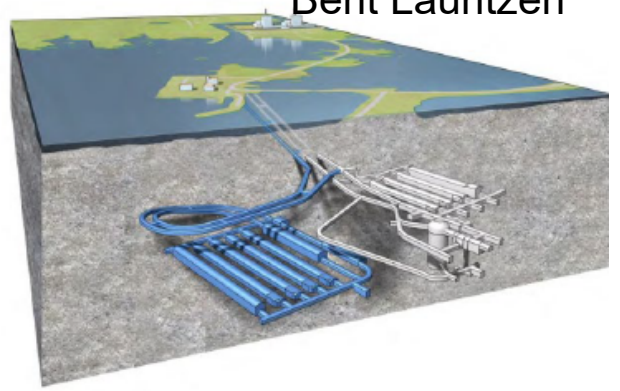
Spent fuel – typical composition

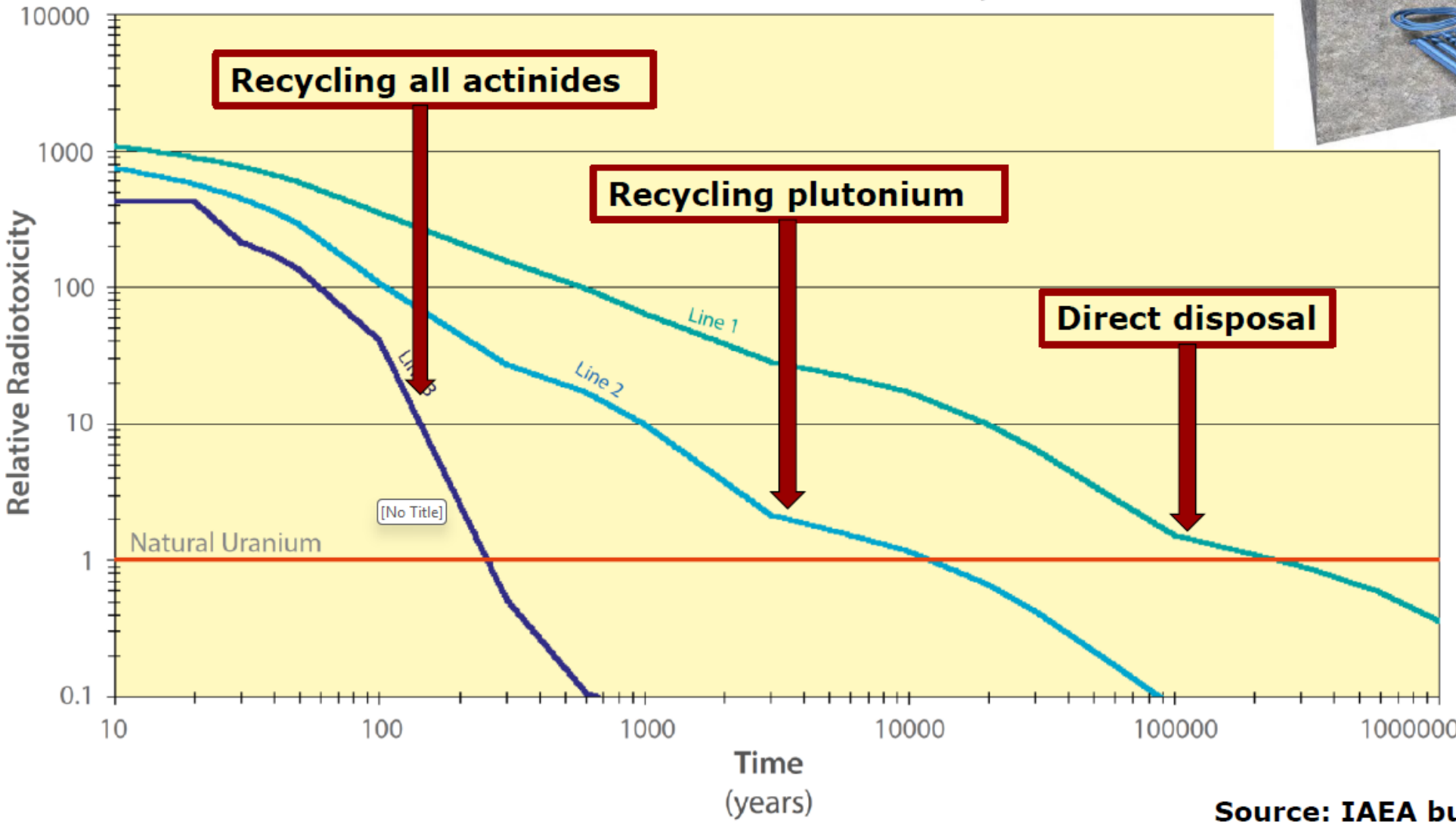
	Radionuclide	Half-Life [a]	Mass [kg]	Activity [TBq]
Remainder of original uranium fuel	^{235}U	7.04×10^{8}	209.0	1.67×10^{-2}
	^{238}U	4.46×10^{9}	22,141	0.28
	Total	------	**22,350**	**0.30**
Actinides (transuranium)	^{234}U	2.45×10^{5}	6.2	1.43
	^{236}U	2.34×10^{7}	137.5	0.33
	^{238}Pu	87.7	5.6	3.55×10^{3}
	^{239}Pu	2.41×10^{4}	177.2	4.07×10^{2}
	^{240}Pu	6.56×10^{3}	69.1	5.81×10^{2}
	^{241}Pu	14.3	37.2	1.43×10^{5}
	^{242}Pu	3.75×10^{5}	15.3	2.23
	^{237}Np	2.14×10^{6}	13.5	0.352
	^{241}Am	4.33×10^{2}	6.8	0.86×10^{3}
	others	-------	3.9	------
	Total	-------	**472.3**	$\mathbf{1.48 \times 10^{5}}$
Fission products	^{89}Sr	0.138	0.1	1.08×10^{5}
	^{90}Sr	28.9	18.0	0.91×10^{5}
	^{106}Ru	1.02	6.7	0.82×10^{6}
	^{134}Cs	2.05	35.1	1.69×10^{6}
	^{137}Cs	30.0	42.9	1.38×10^{5}
	^{144}Ce	0.78	13.8	1.62×10^{6}
	^{147}Pm	2.62	12.0	4.12×10^{5}
	^{154}Eu	8.59	1.2	1.20×10^{4}
	Others	-------	136.3	0.85×10^{5}
	Stables	-------	910.5	-------
	Total	-------	**1176.6**	$\mathbf{49.8 \times 10^{5}}$
	Grand Total	-------	**23,9989**	$\mathbf{51.3 \times 10^{5}}$

Typical composition of spent fuel after 1 GW production for a year. (*From Terranova et al. J. Nucl. Eng. 2024, 5, 299–317*).

Radioactive waste management

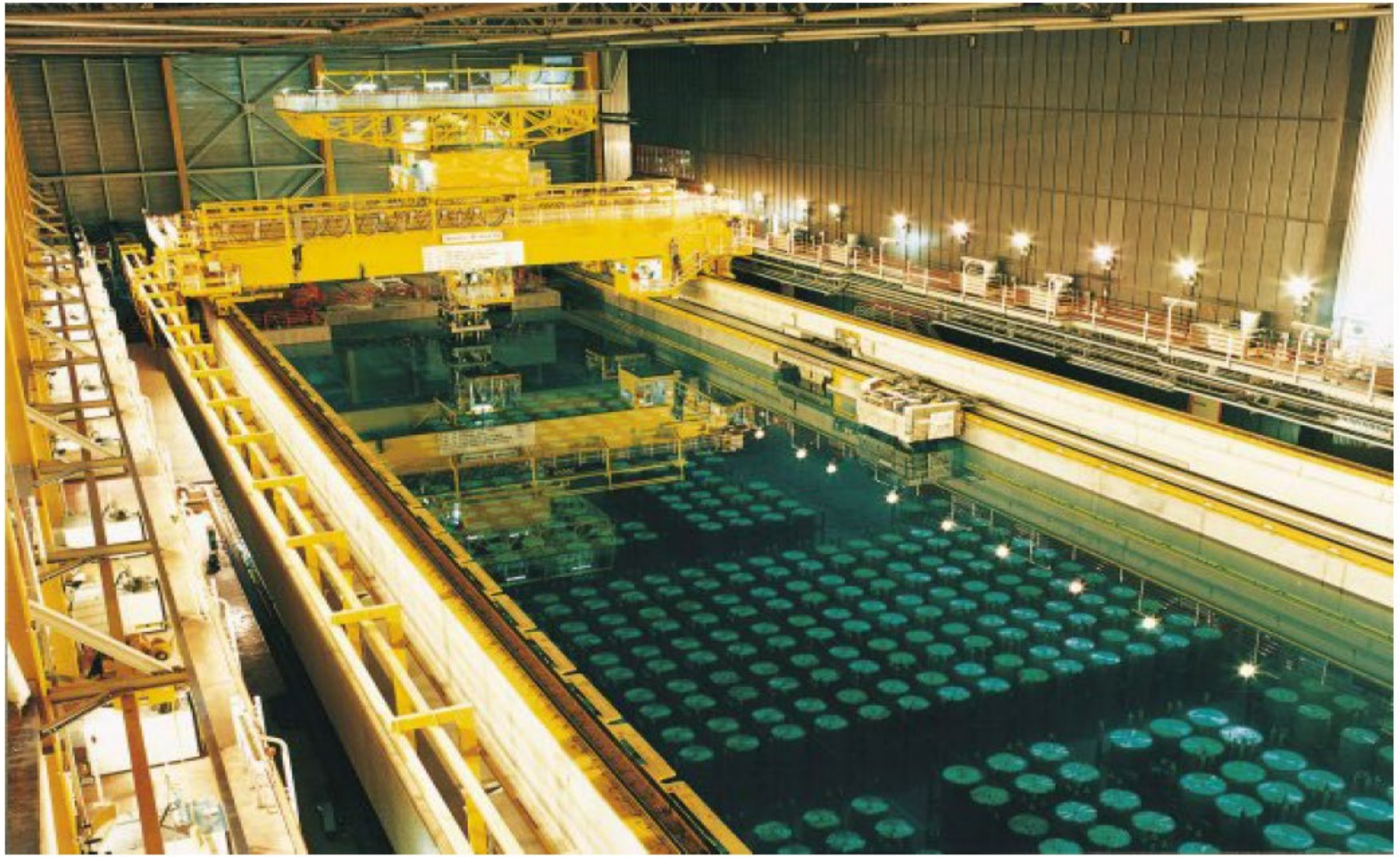
Bent Lauritzen





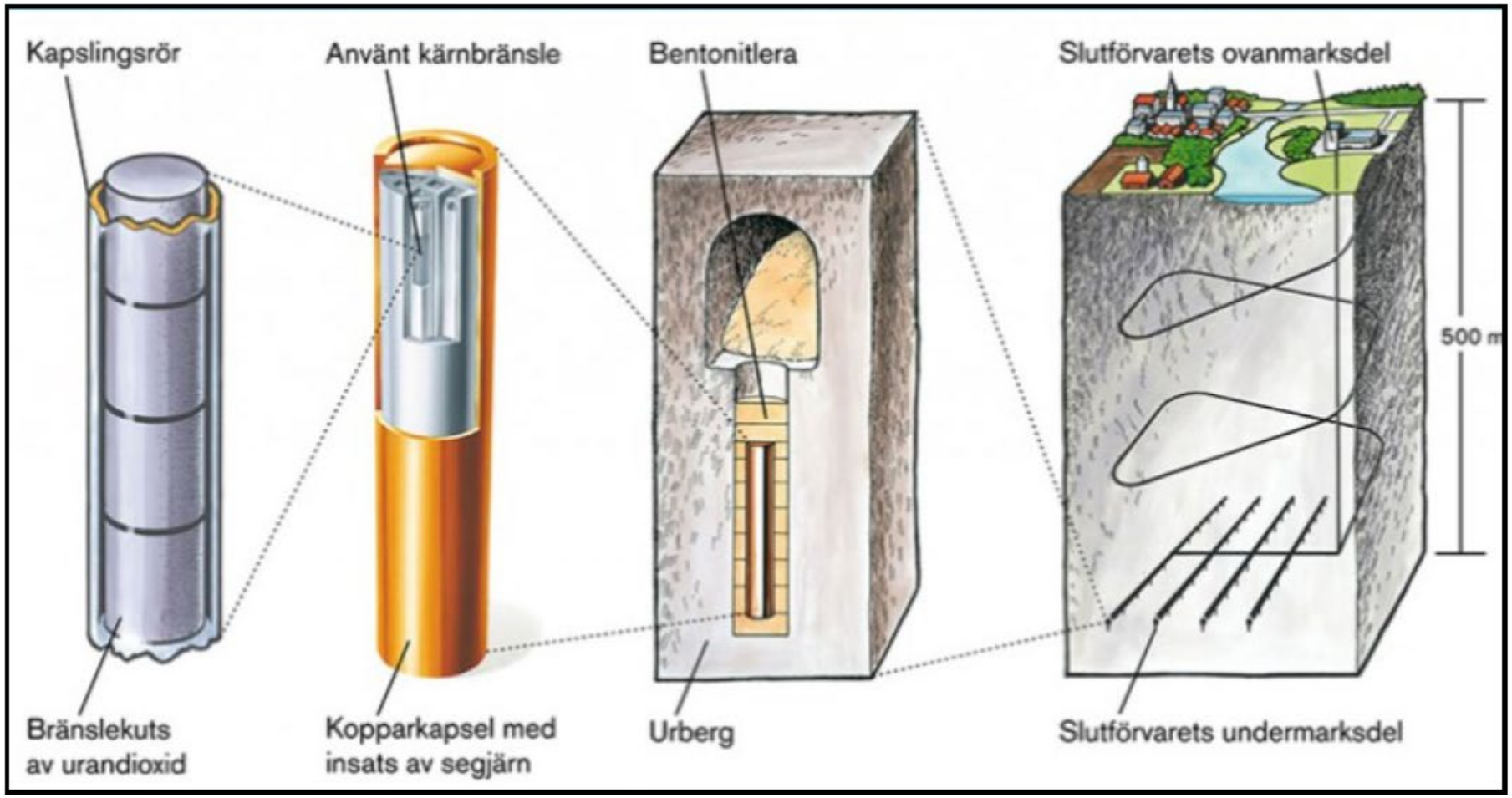
Source: IAEA bulletin 55, 2014

Radioactive waste management



Storage pond for used fuel at Sellafield, UK

SKB: Kärnbränsleförvaret

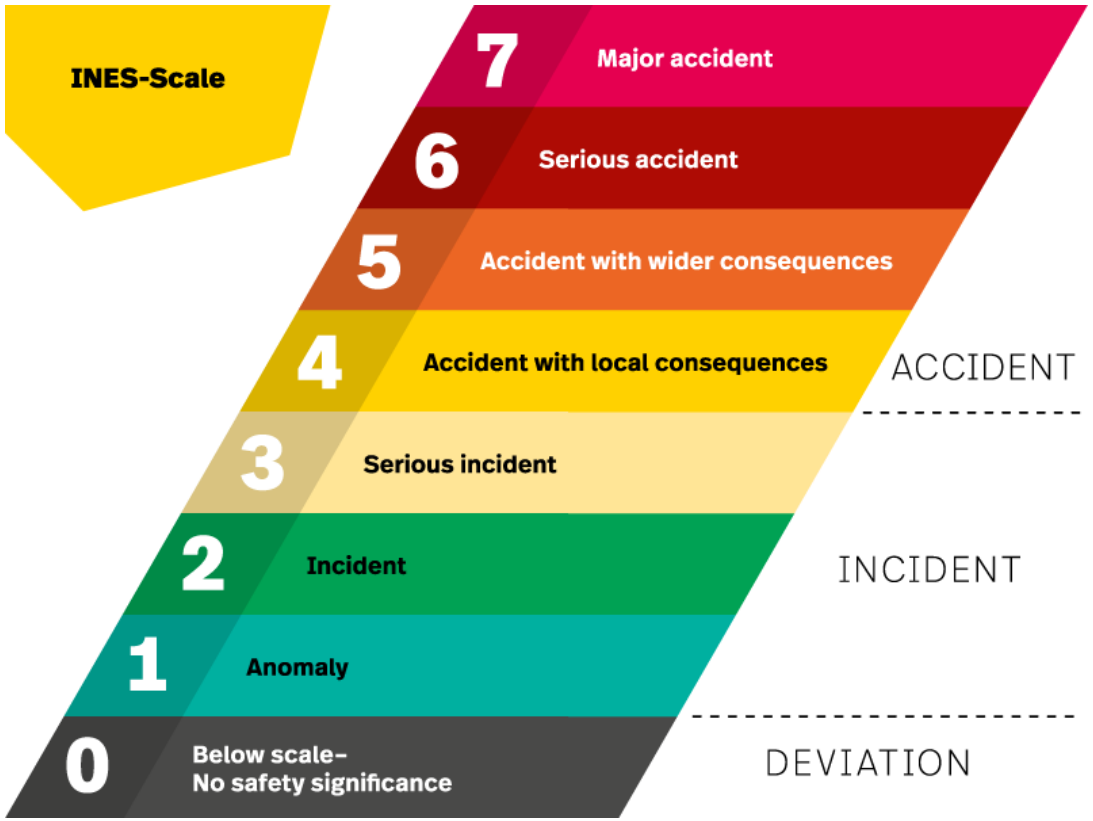


Source: SKB 2022: Vår metod - SKB

Nuclear accidents

The INES scale

(International Nuclear and Radiological Event Scale).



Three Mile Island (USA) 1979, INES 5

- Mechanical error on valve + human errors
- Partly meltdown
- Minor radioactive release
- No evacuation

Chernobyl (Ukraine / USSR) 1986, INES 7

- Failed safety test, explosion and graphite fire
- Massive radioactive release
- Evacuation of 350,000 people

Fukushima Daiichi (Japan) 2011, INES 7

- Earthquake cut off external power
- Tsunami disabled backup power
- Hydrogen explosion + meltdown in three reactors
- Nearly half a million people evacuated

Future directions

Generation IV fission reactors

Objectives with gen. IV reactors

- Better utilization of the fuel (breeding, Pu cycle, Th cycle)
- Higher temperature (thermal efficiency, process heat) – other coolants
- Better economy
- Higher safety
- Non-proliferation goals

Main classes

Super-Critical Water Reactor (SCWR),

Sodium-cooled Fast Reactor (SFR),

Gas-cooled Fast Reactor (SFR),

Lead-cooled Fast Reactor (LFR),

Very-High Temperature Reactor (VHTR),

Molten Salt Reactor (MSR).

Reactor fuel cycles

- **Fission of ^{235}U in thermal reactors (once through):**

$$n + {}^{235}U \rightarrow \text{fission fragments} + n + n + \ldots \quad \textbf{(0.7\%)}$$

- **Fuel breeding in fast reactors (closed cycle, plutonium):**

$$n + {}^{238}U \rightarrow {}^{239}U \rightarrow {}^{239}Np \rightarrow {}^{239}Pu$$

$$n + {}^{239}Pu \rightarrow \text{fission fragments} + n + n + \ldots \quad \textbf{(99.3\%)}$$

- **Thorium fuel cycle?**

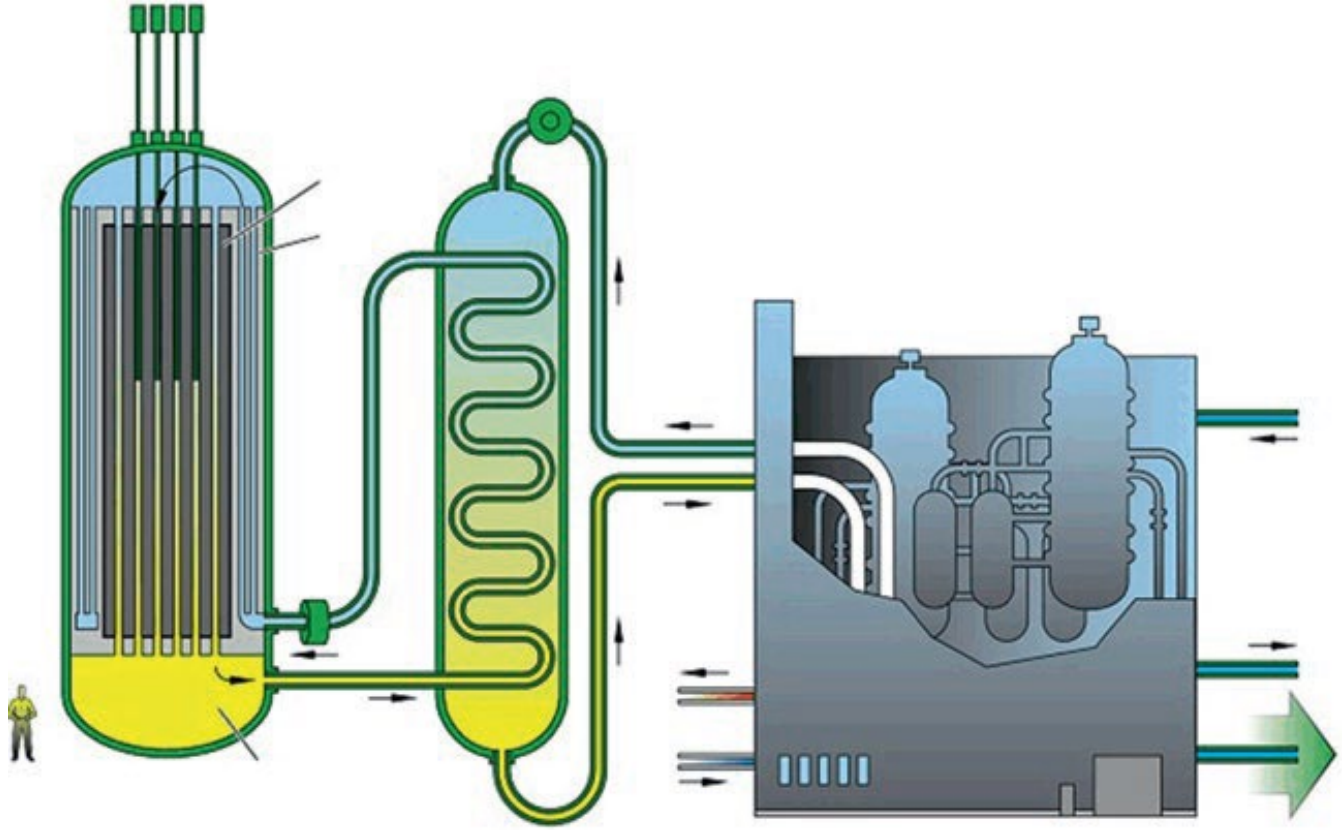
$$n + {}^{232}Th \rightarrow {}^{233}Th \rightarrow {}^{233}Pa \rightarrow {}^{233}U$$

$$n + {}^{233}U \rightarrow \text{fission fragments} + n + n + \ldots$$

Very-High-Temperature Reactor (VHTR)

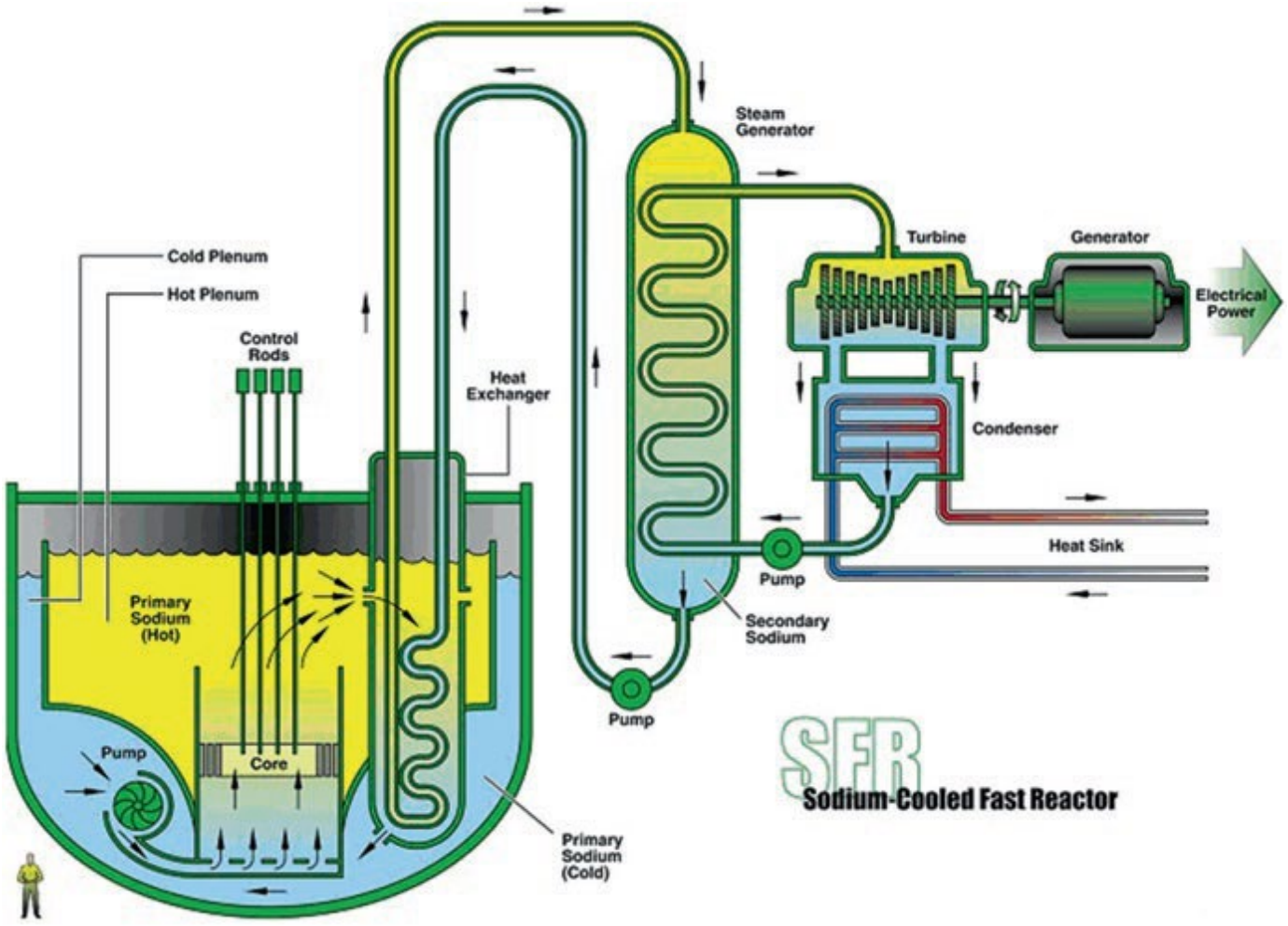
Or High-temperature gas cooled reactor (HTGR)

- He-cooled with thermal neutrons
- Graphite moderated
- Thermal neutrons
- SiC or ZrC coated fuel elements
- Reactor temperature up to 1600 °C
- Coolant out: up to 950 °C
- High thermal efficiency



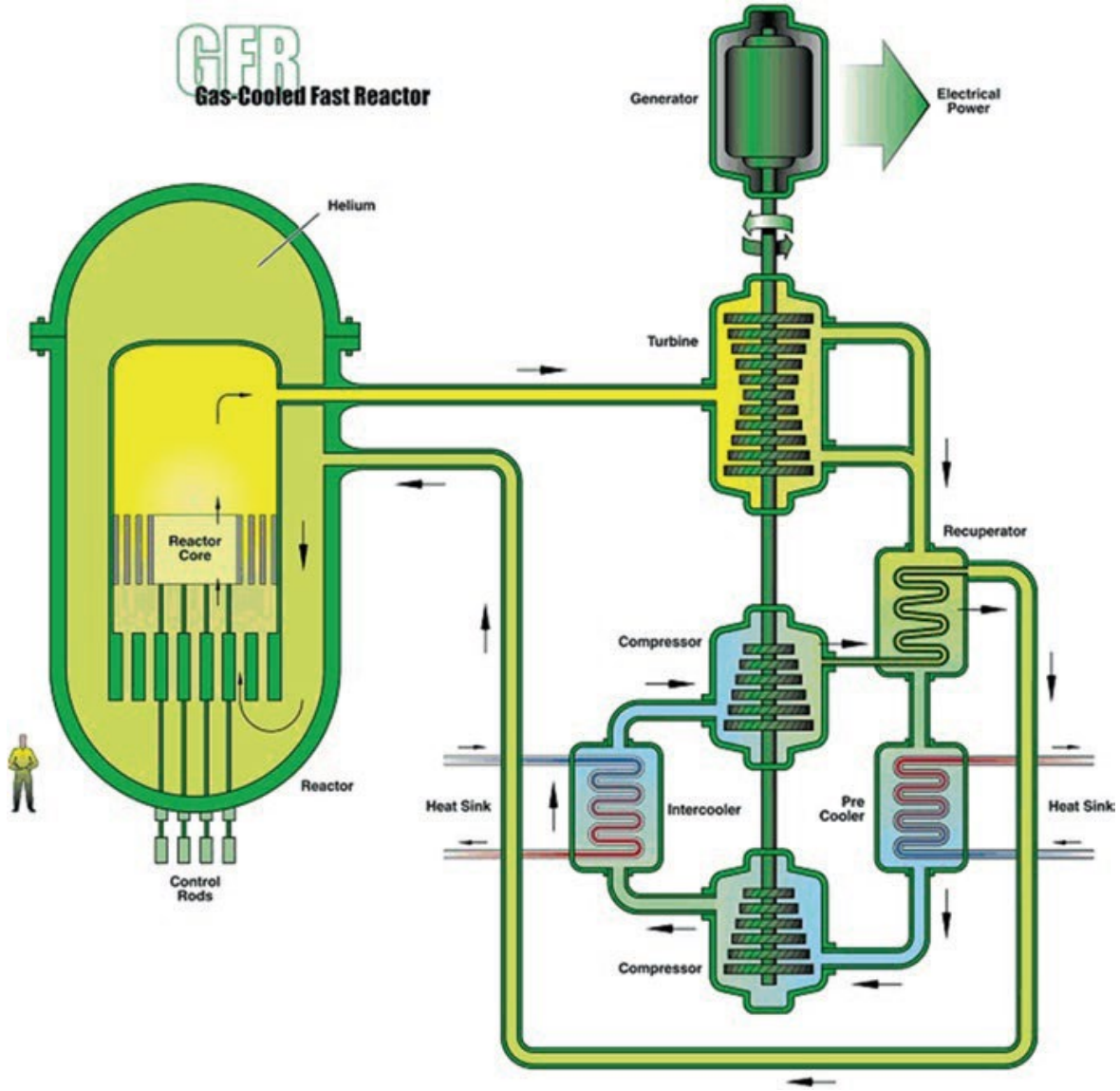
Sodium-Cooled Fast Reactor (SFR)

- Or Liquid Metal Fast Reactor
- Liquid Sodium or Pb-Bi as coolant
- High boiling point
- Low pressure
- Fast neutrons
- Liquid metal does not moderate much
- Breeder, converts ^{238}U ant Th.
- Coolant out: Na: 550 °C
- Coolant out: Pb-Bi: 550 °C



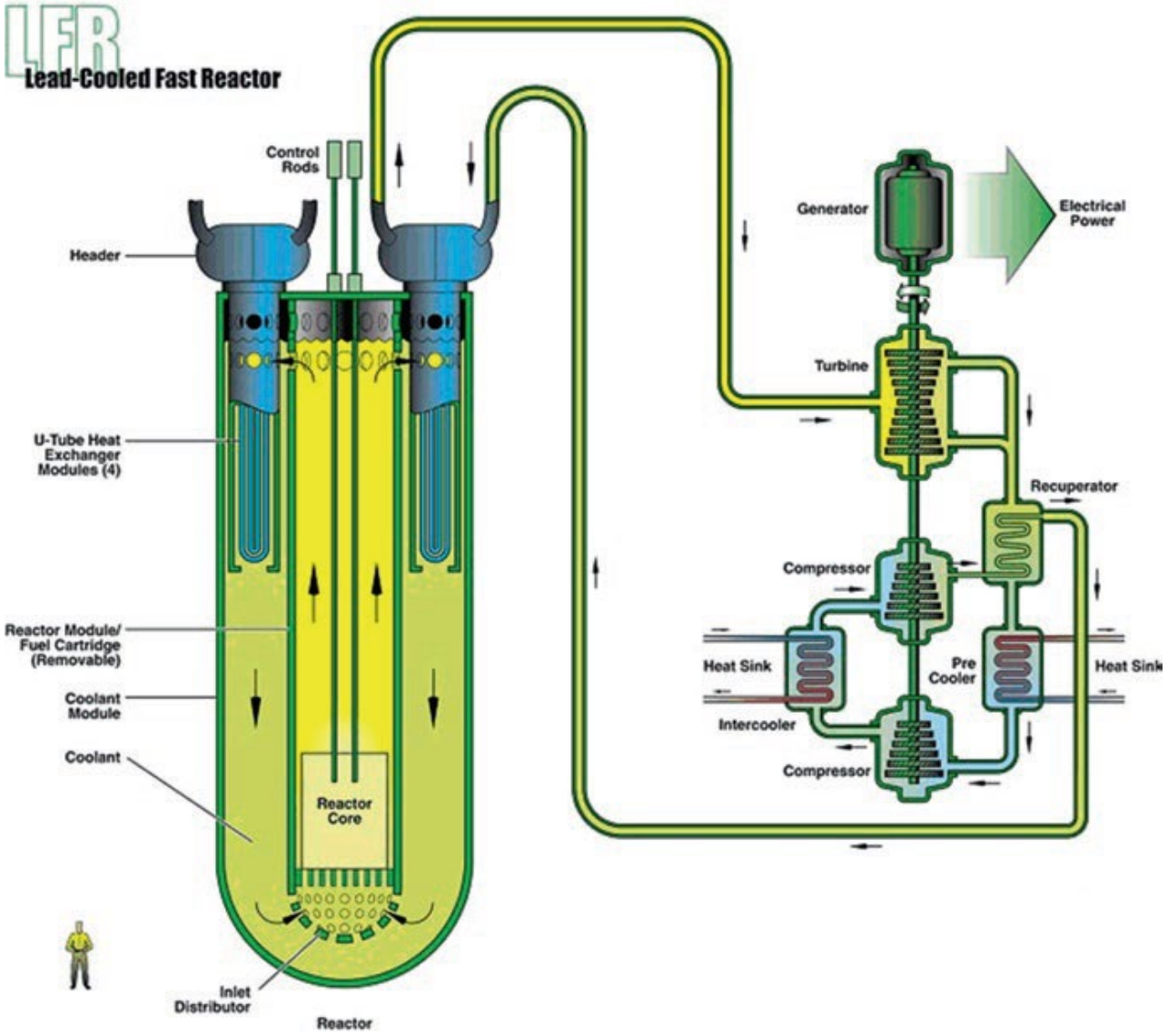
Gas-Cooled Fast Reactor (GFR)

- Similar to High-Temperature gas cooled reactor (HTGR)
- But with fast neutrons



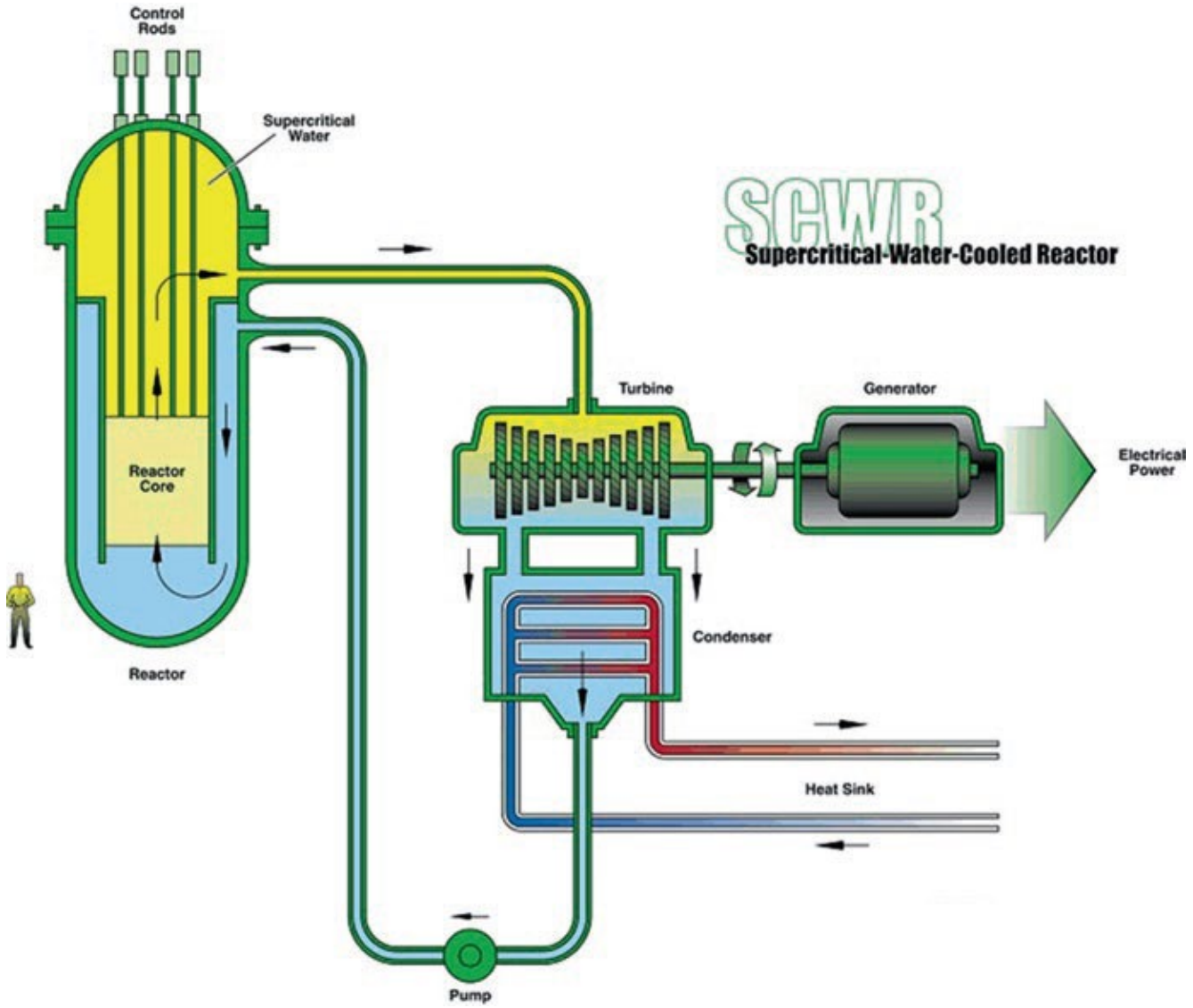
Lead-Cooled Fast Reactor (LFR)

- Another liquid metal cooled reactor.
- Fast neutrons



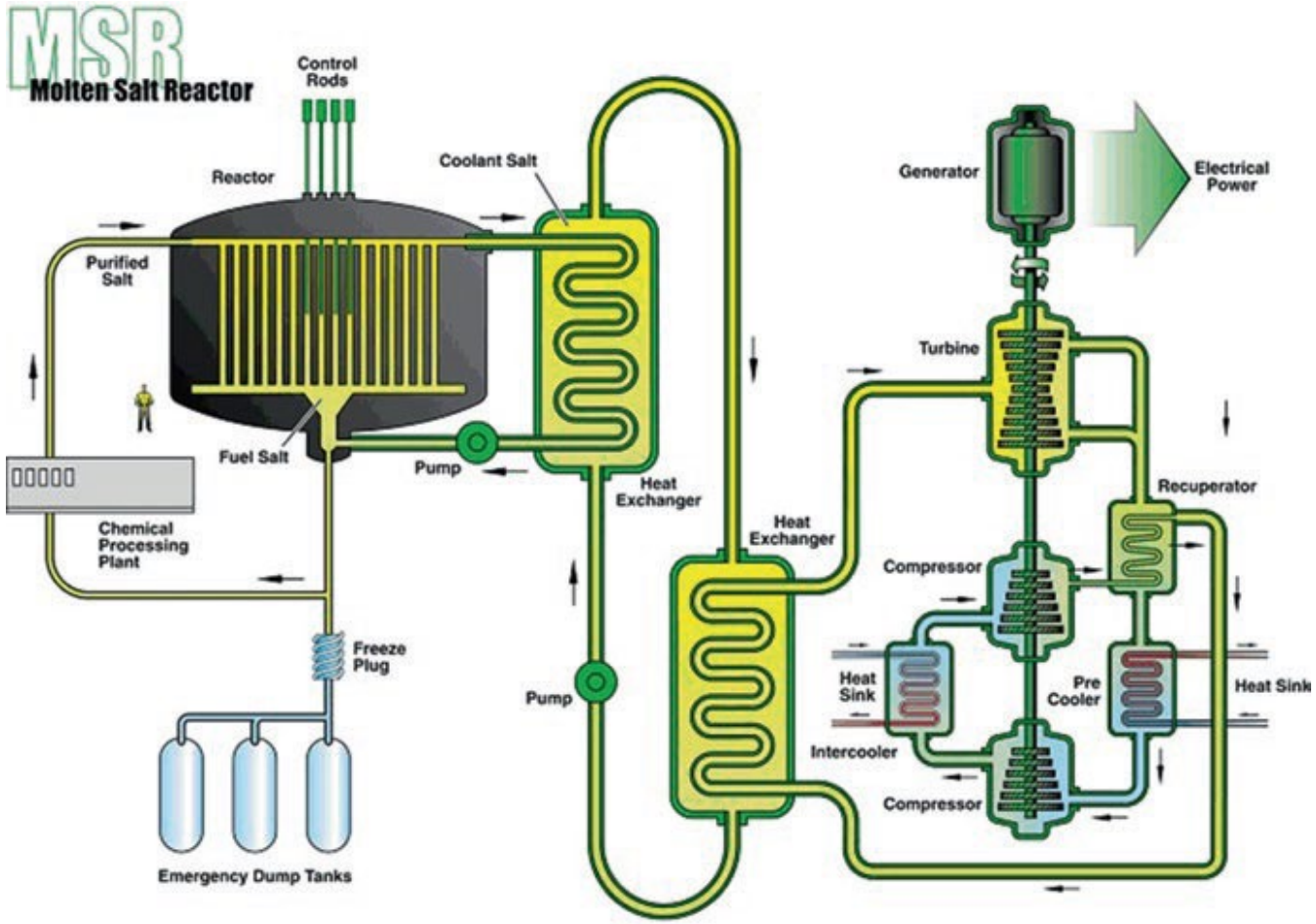
Supercritical Water-Cooled Reactor (SCWR)

- Operate above the thermodynamic critical point of water (374 °C, 22.1 MPa)
- Thermal or fast neutrons
- Coolant out up to 625 °C



Molten Salt Reactor (MSR)

- Originally developed in the 1950s and 1960s in the United States (Oak Ridge National Laboratory).
- Fuel as salts dissolved in a molten fluoride salt.
- Fuel mix and coolant same thing
- Fuel flexibility
- Thermal neutrons
- Addition of nuclear waste
- Fuel can be replaced
- Limited amount of fuel at a time
- I case of accident, fuel drained and cooled
- Coolant out 750-1000 °C



Overview of IV gen. reactors

Table 1.1 Summary of the main characteristics of the six Generation IV systems

System	Neutron spectrum	Coolant	Temp. °C	Fuel cycle	Size (MWe)
VHTR (very-high-temperature reactor)	Thermal	Helium	900–1000	Open	250–300
SFR (Sodium-cooled fast reactor)	Fast	Sodium	550	Closed	30–150 300–1500 1000–2000
SCWR (Supercritical water-cooled reactor)	Thermal/fast	Water	510–625	Open/closed	300–700 1000–2000
GFR (Gas-cooled fast reactor)	Fast	Helium	850	Closed	1200
LFR (Lead-cooled fast reactor)	Fast	Lead	480–800	Closed	20–180 300–1200 600–1000
MSR (Molten salt reactor)	Epithermal	Fluoride salt	700–800	Closed	1000

Bahman Zohuri, Small Modular Reactors as Renewable Energy Sources. Springer Nature 2019

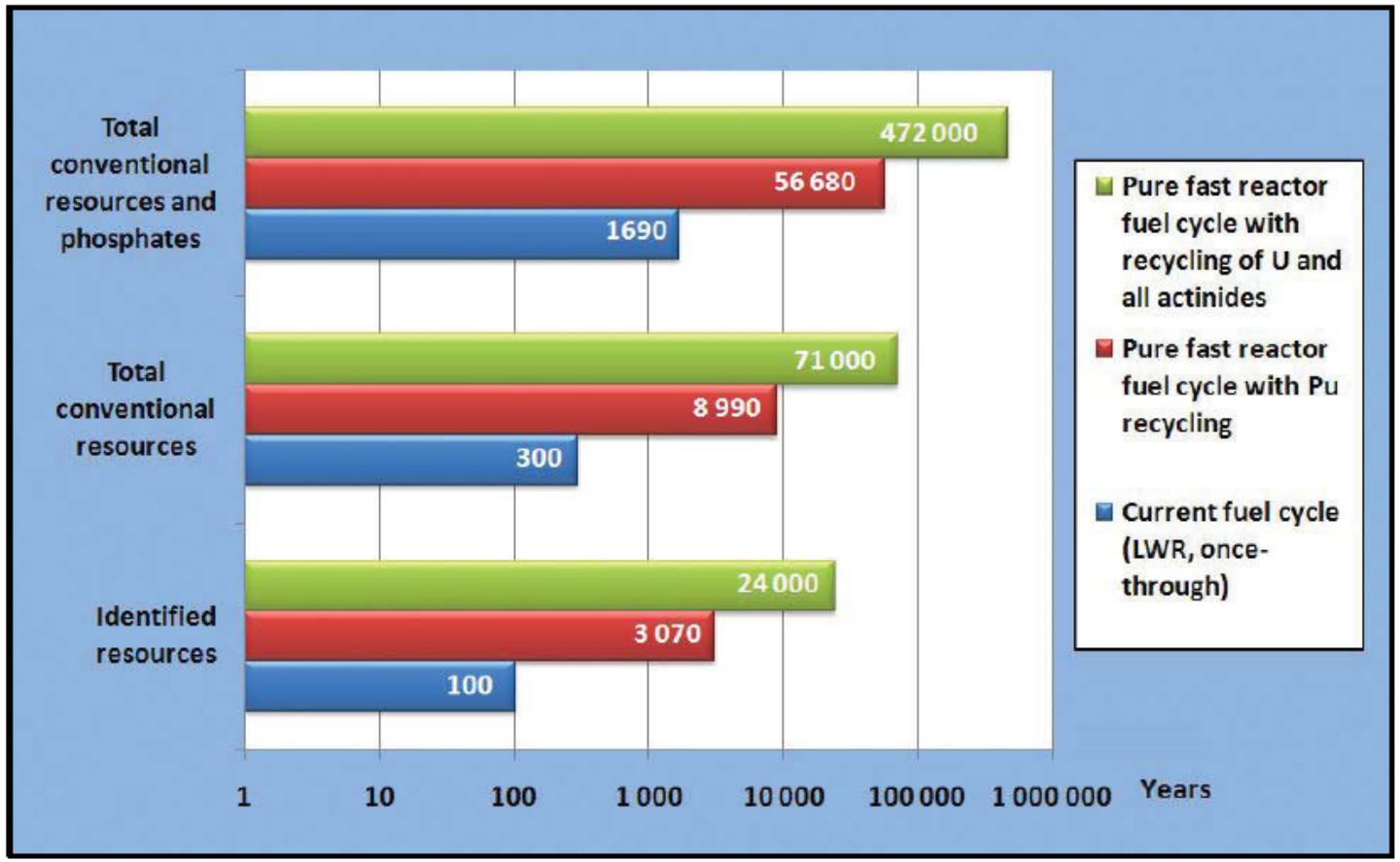
TABLE 2.2

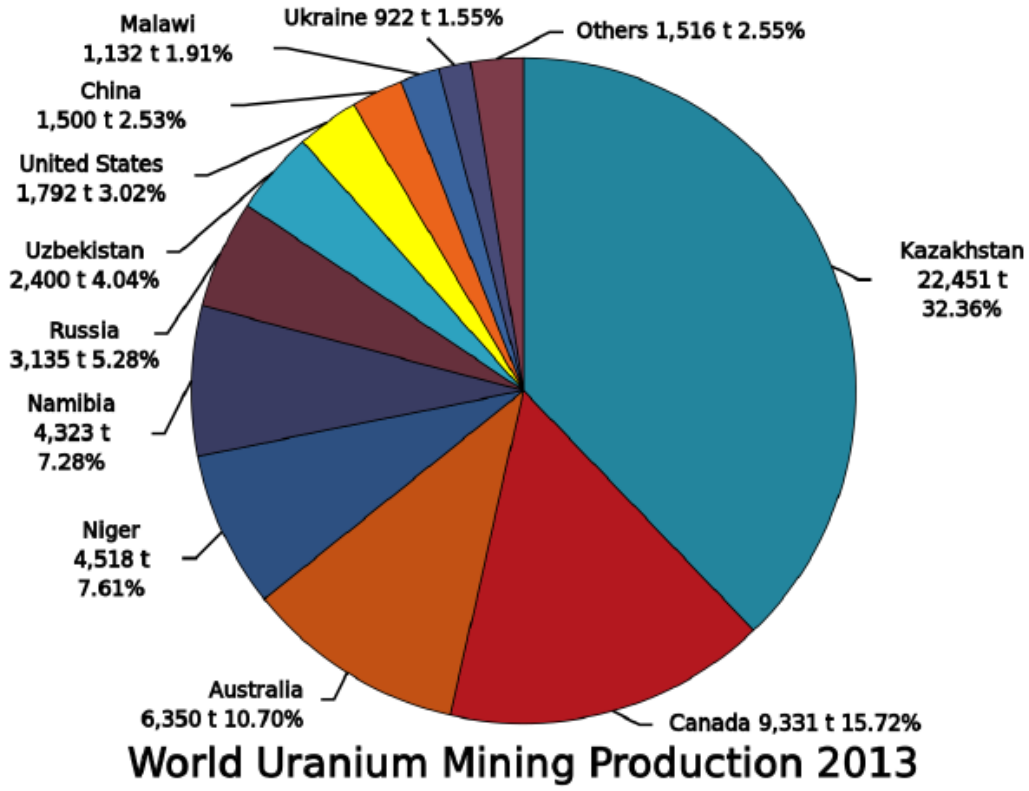
Summary of Nuclear Reactor Parameters for Hydrogen Production

	Neutron Spectrum	Fuel Cycles being Used or Studied	Coolant	Reactor Outlet Coolant Temperature (°C)	Typical Sizes MWt/MWe	Electricity Generation Route
Light-water reactors: PWR, AP, EPR; BWR ABWR	Thermal	UO_2; U–Pu MOX	Light water	280–325	2000–4800 MWt 600–1700 MWe	Steam turbine
Heavy-water reactors: CANDU, ACR	Thermal	Natural UO_2; low enriched UO_2;	Heavy water	310–319	2000–3200 MWt; 700–1100 MWe	Steam turbine
Supercritical water reactor: S–LWR, CANDU SCWR, SF–LWR	Thermal or fast	UO_2; Th/U; U–Pu MOX	Light water	430–625	1600–2540 MWt 700–1150 MWe	Steam turbine
Liquid metal fast reactors: SFR, LFR	Fast	U–Pu MOX; U–Pu nitrides; MOX w/TRU	Sodium; lead; lead bismuth	SFR: 550 Pb or Pb–Bi: 500–800	45–3000 MWt 20–1100 MWe	Steam turbine; SCO_2 turbine
Molten salt reactors: AHTR	Thermal	UO_2; Th/U	Salts: $^7Li_2BeF_4$ $NaF–ZrF_4$ and so on.	750–1000	900–2400 MWt 400–1200 MWe	Steam turbine; SCO_2 turbine
Gas-fast reactor: GFR	Fast	U–Pu MOX; U–TRU	Helium	850	600–2400 MWt 280–1100 MWe	Steam turbine; gas turbine
High-temperature gas reactors: HTGR, VHTR	Thermal	UO_2; Th/U; PuO_2; U–Pu MOX; U–TRU	Helium	750–950	100–600 MWt 45–300 MWe	Steam turbine; gas turbine

Xing L. Yan, Ryutaro Hino eds. Nuclear Hydrogen Production Handbook. CRC Press, Taylor & Francis Group 2011

Nuclear – Security of supply





World Uranium Mining Production 2013

Abundant uranium resources with fast reactor technology
More than 100 000 years!

Sources: IAEA, DOE

Small Modular Reactors – close to market

European SMR options



Water-cooled



U.S. options considered for deployment in EU



NuScale (Water cooled)

- Interests expressed by Bulgaria, Czech Republic, Poland, Romania & Ukraine.



BWRX-300, GE-Hitachi (Water cooled)

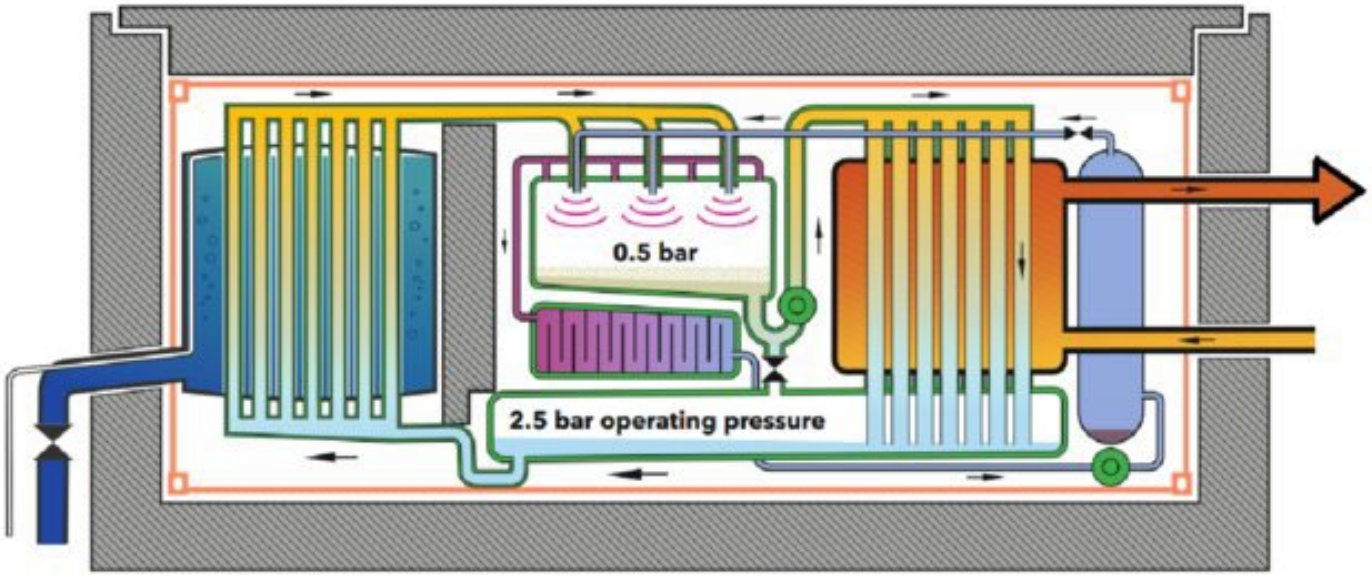
- Interests expressed by with Czech Republic, Estonia & Poland

Source: European Commission, 2022

Copenhagen Atomics (DK startup)

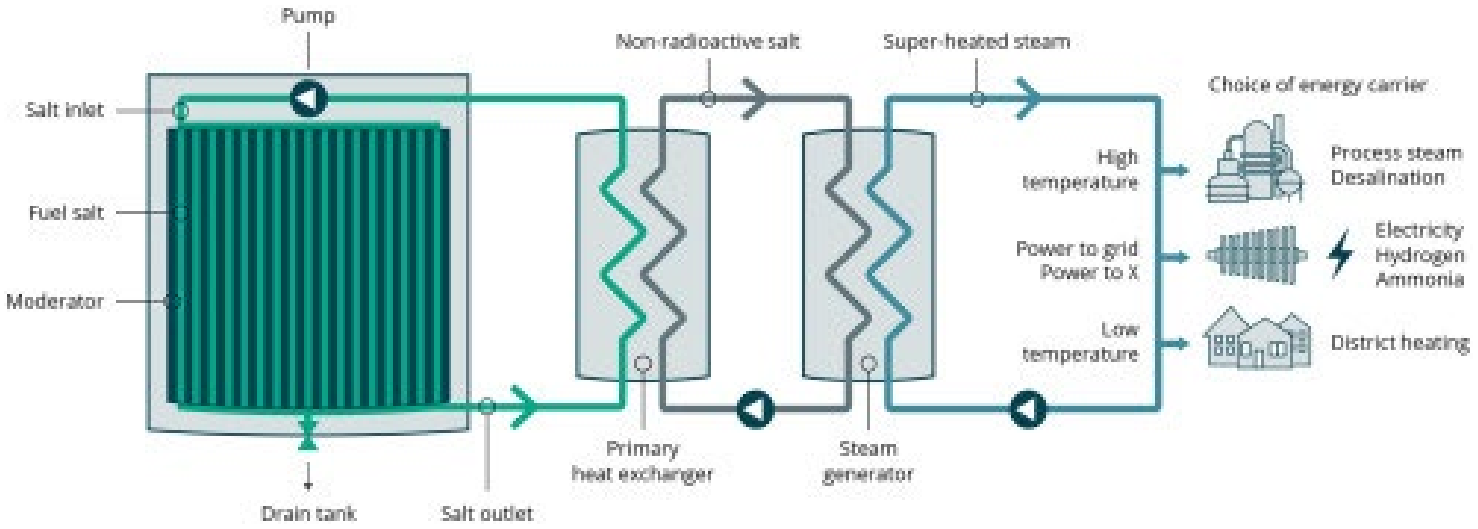




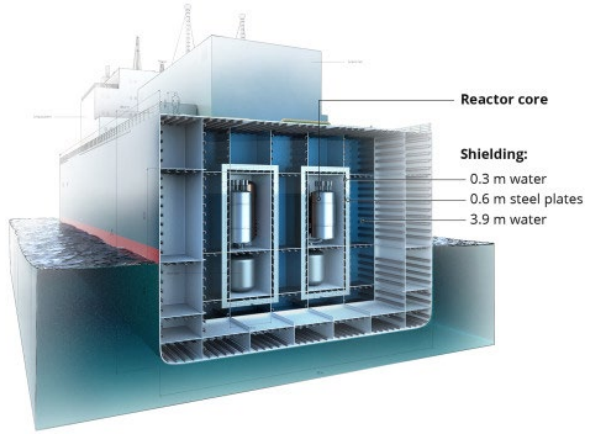


- ”Wasteburner” MSR
- Container solution
- Thermal Molten Salt Reactor
- Fluoride salts, U-Th fuel
- D_2O moderator
- 250 MW_{th}/ 100 Mw_e

Seaborg Technologies (DK startup)- now Saltfoss



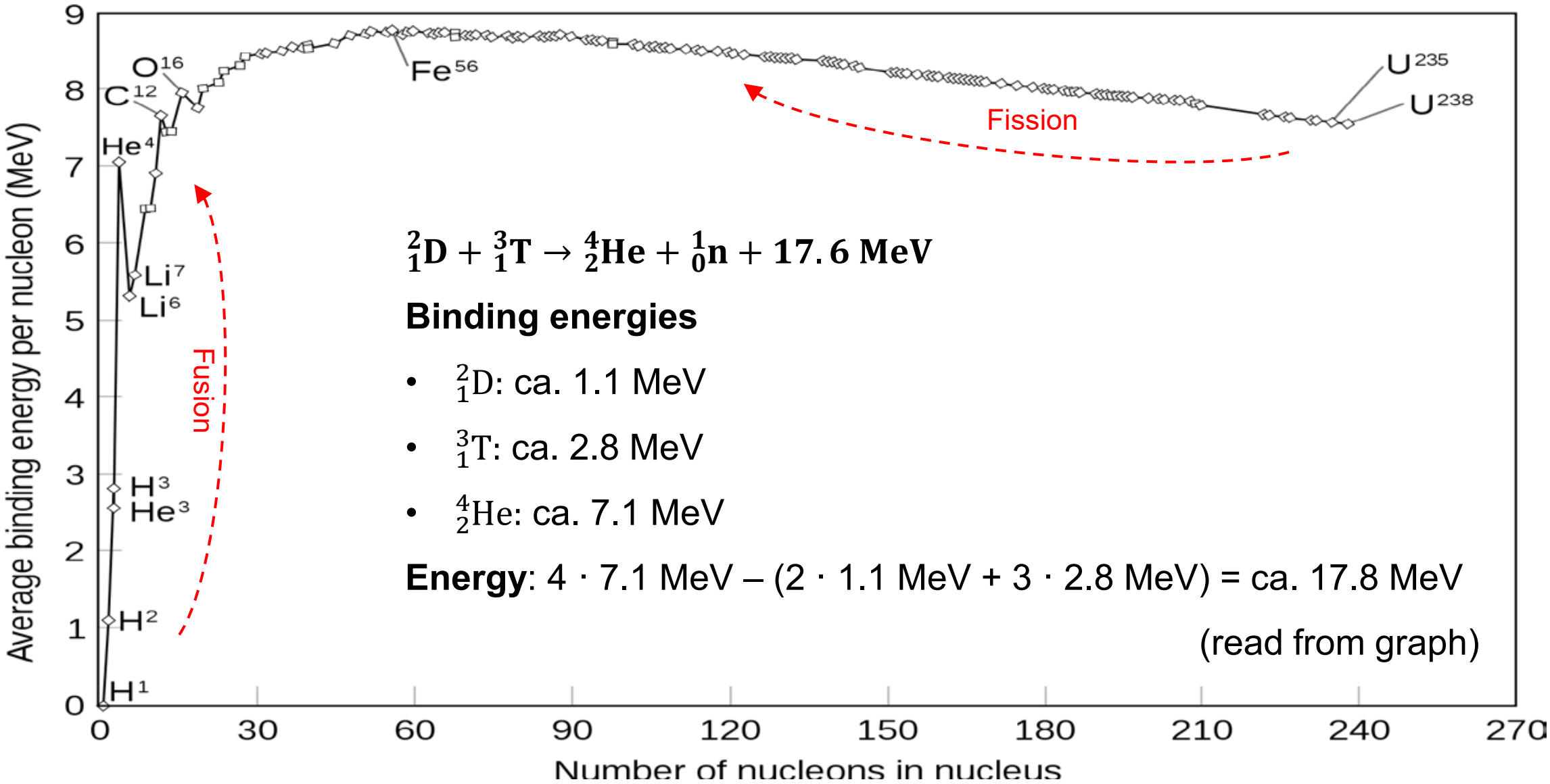




- Thermal Molten Salt Reactor
- Fluoride salts, U fuel,
- Graphite moderator
- 250 MW_{th}/ 100 MW_e
- 12 year fuel cycle

Fusion

Binding energy and mass defect - again



$$^{2}_{1}\mathrm{D} + {}^{3}_{1}\mathrm{T} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

Binding energies

- $^{2}_{1}\mathrm{D}$: ca. 1.1 MeV
- $^{3}_{1}\mathrm{T}$: ca. 2.8 MeV
- $^{4}_{2}\mathrm{He}$: ca. 7.1 MeV

Energy: 4 · 7.1 MeV – (2 · 1.1 MeV + 3 · 2.8 MeV) = ca. 17.8 MeV

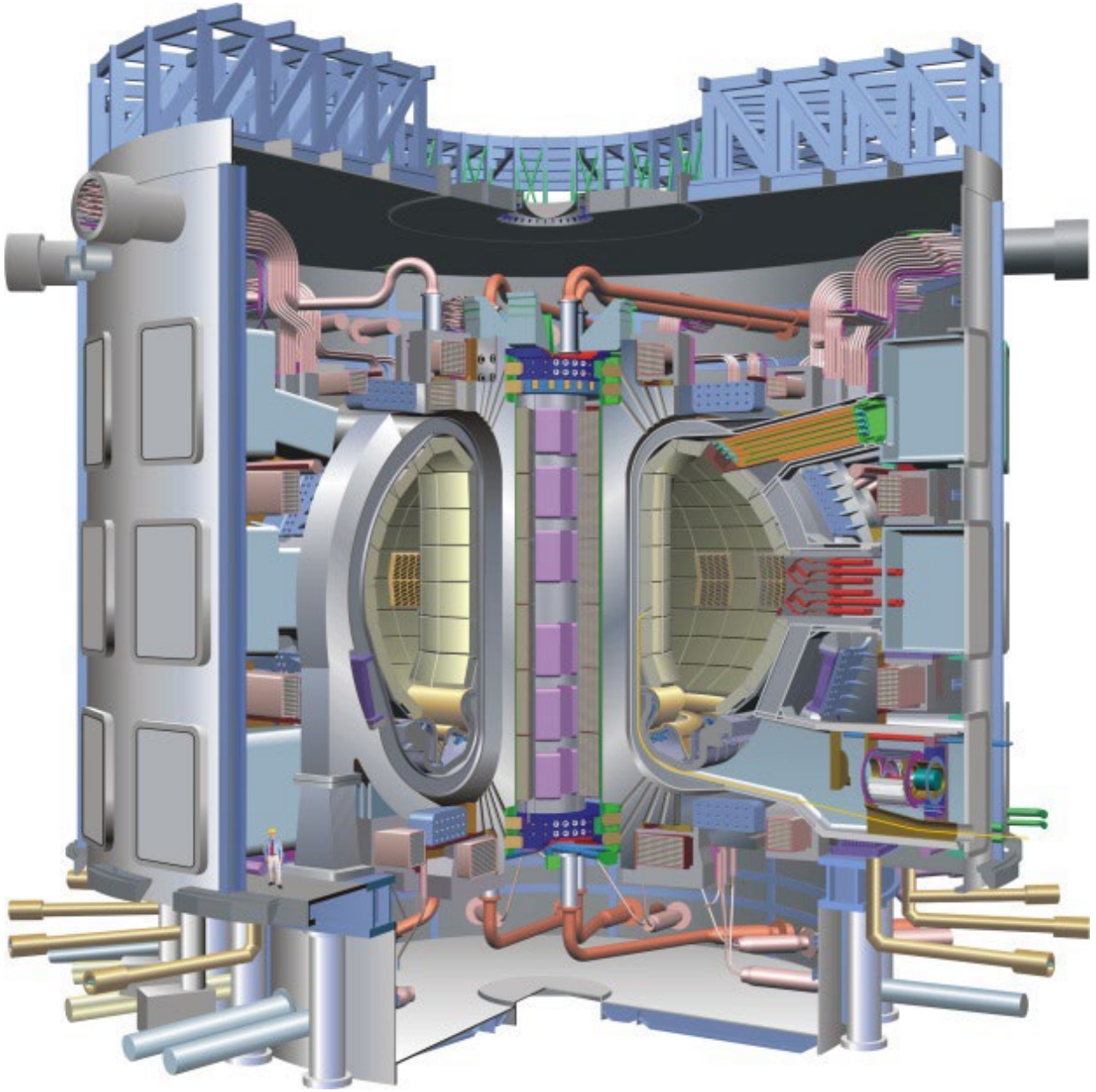
(read from graph)

Fusion

Fusion of deuterium and tritium to Helium

$$^{2}_{1}\mathrm{D} + ^{3}_{1}\mathrm{T} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

- Not neutron initiated
- Nuclei brought together by collisions
- Extreme temperatures (+ 100 mil. Degrees)
- Plasma maintained in magnetic field
- D and T have highest cross section for fusion

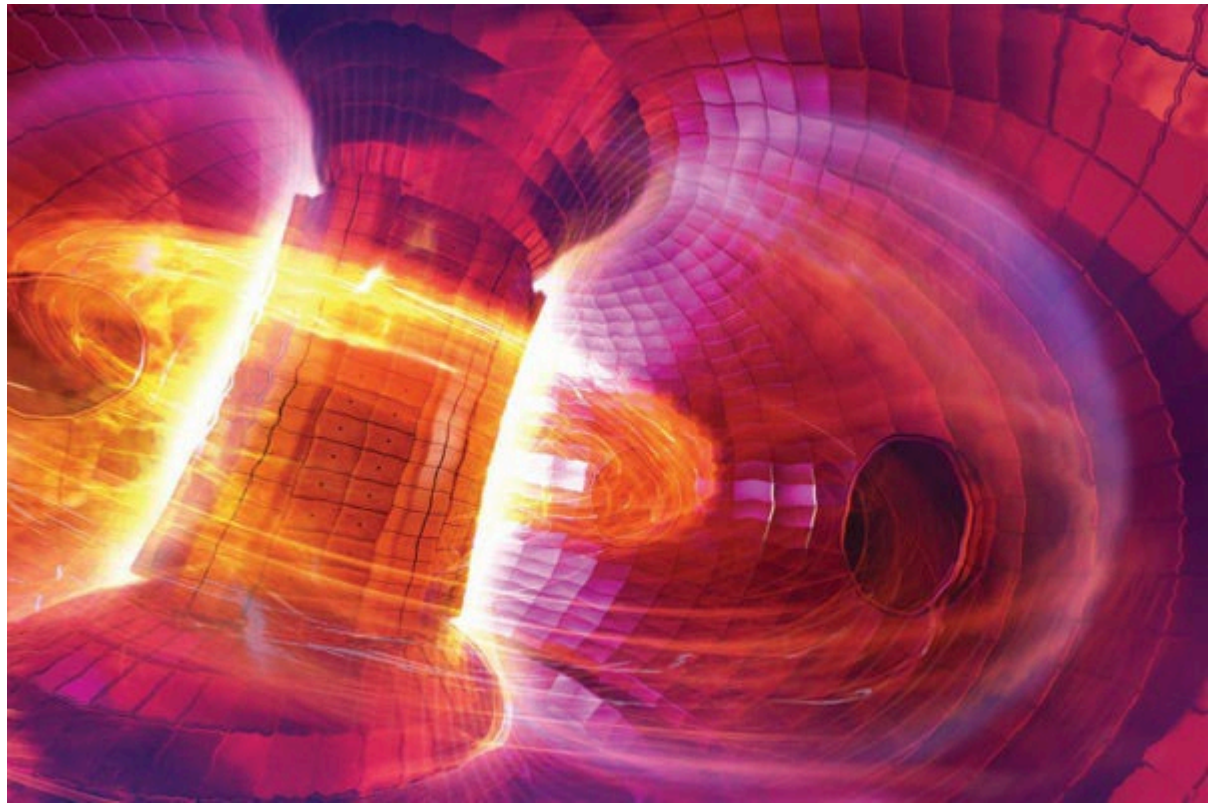


International Thermonuclear Experimental Reactor, Iter, France

Fusion – harvesting the energy

$$^{2}_{1}\mathrm{D} + ^{3}_{1}\mathrm{T} \rightarrow ^{4}_{2}\mathrm{He} + ^{1}_{0}\mathrm{n} + 17.6\ \mathrm{MeV}$$

Carriers	Energy
Neutrons	14.1 MeV
Alpha paricles	3.5 MeV
Total	17.6 Mev



Most energy carried by neutrons

No charge – not affected by plasma

Penetrate metal

Collected (slowed) in a “blanket” outside

Finally absorbed by Lithium forming tritium

$$^{1}_{0}\mathrm{D} + ^{6}_{3}\mathrm{Li} \rightarrow ^{3}_{1}\mathrm{T} + ^{4}_{2}\mathrm{He}$$


That's all Folks!
SUPERB wallpapers